Drought-Induced Deforestation in the Brazilian Amazon*

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Drought in the Amazon has become more frequent and intense, contributing to more forest losses in recent years. This paper examines drought-induced deforestation in the Brazilian Legal Amazon and investigates the economics behind it. We focus on agricultural drought to study differential roles of family versus commercial farms in this region. Using high-resolution root-zone soil moisture and land cover data, we document that increases in soil moisture deficits lead to higher deforestation within the same year, with no detectable within-year feedback from local deforestation to local rainfall once we condition on prior forest stock. Our heterogeneity analyses by family-farm area shares across 770 municipalities reveal the critical role of resource-restricted family farms: (1) drought only raises deforestation where family farming predominates, and not where commercial agriculture dominates; (2) forest-related environmental fines rise rather than fall in high-family-farm municipalities during droughts, ruling out weaker monitoring; (3) high-family-farm municipalities see staple output fall, credit contract, and irrigation fail to expand, while low-family-farm municipalities suffer large cattle losses. Our estimates show that agricultural drought accounts for roughly 331 square kilometers of primary forest clearing during 2006-2019, committing carbon damages of about $135 million per year.

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1 — Introduction

How do climate shocks reshape land-use decisions at tropical forest frontiers? The question lies at the intersection of the economics of climate change, agricultural development, and environmental conservation because the world's remaining tropical forests are simultaneously major carbon sinks and active margins of agricultural expansion. The Brazilian Legal Amazon exemplifies this tension. It stores one of the largest stocks of terrestrial carbon on Earth while supporting a rapidly evolving agricultural frontier whose land-use decisions are shaped by commodity markets, environmental regulation, and rural development policies (Hanusch 2023). Although Brazil has substantially reduced deforestation through satellite monitoring, law enforcement, supply-chain agreements, and restrictions on rural credit (Gibbs et al. 2016; Souza-Rodrigues 2019; Heilmayr et al. 2020; Baragwanath and Bayi 2020; Assunção et al. 2020; Assunção, Gandour, and Rocha 2023; Baragwanath and Shinde 2025), climate variability has emerged as an increasingly important and comparatively understudied determinant of forest loss (Araujo et al. 2023; Araujo, Costa, and Sant'Anna 2025; Hsiao et al. 2026).

Drought is particularly important because it is becoming both more frequent and more severe across the Amazon basin. Recent ecological evidence documents increasingly intense soil-moisture deficits and more frequent megadroughts under continued climate change (Anderson et al. 2018; Chen et al. 2023). Existing research has primarily examined either the ecological consequences of drought or the reverse relationship through which deforestation alters regional rainfall and moisture recycling (Staal et al. 2020; Smith, Baker, and Spracklen 2023; Papastefanou et al. 2024). Much less is known about how agricultural drought affects the economic incentives of producers operating at the forest frontier, and in particular whether forest clearing itself constitutes a behavioral response to drought shocks rather than solely a driver of them.

Investigating this question requires understanding how producers respond to drought under different economic environments. In the standard land-use model, forest is cleared whenever the expected return from agricultural production exceeds the value of leaving land under forest (Chomitz and Gray 1996; De Janvry, Fafchamps, and Sadoulet 1991; Pfaff 1999; Rudel et al. 2005; Amacher, Ollikainen, and Koskela 2009). A drought lowers agricultural

productivity and expected farm profits, increasing the relative opportunity cost of converting forest into cropland or pasture. Under this logic, drought should reduce deforestation rather than increase it. Consistent with this prediction, a large literature shows that changes in agricultural productivity alter land-use decisions by changing the profitability of cultivation (Abman and Carney 2020; Carreira, Costa, and Pessoa 2024; Szerman et al. 2024).

Yet a growing empirical literature documents the opposite relationship. Repeated rainfall deficits increase cropland expansion across developing countries (Zaveri, Russ, and Damania 2020; You et al. 2025), while extreme heat induces persistent forest loss at agricultural frontiers by reducing yields and raising agricultural prices (Hsiao et al. 2026; Khan et al. 2025). Similar evidence emerges from Madagascar, West Africa, and Zambia (Desbureaux and Damania 2018; Leblois 2021; He and Chen 2022; Ordóñez et al. 2025; Tabe-Ojong, Tolani, and Molua 2026). These findings overturn the sign of the textbook prediction but not necessarily the underlying logic: adverse weather raises deforestation because reduced yields, given the inelastic demand for subsistence crops, raise crop prices enough to offset the revenue loss, preserving or increasing the expected profitability of agricultural expansion (Hsiao et al. 2026).

The economics of drought-induced deforestation in the Brazilian Amazon, however, may be more complex: producers in this context differ substantially in their ability to absorb drought shocks. Family and commercial farms operate under markedly different production technologies and financial environments. Commercial producers generally have greater access to formal credit, insurance, irrigation, and diversified production systems, allowing them to smooth temporary income fluctuations. Family farms, by contrast, depend primarily on rain-fed production, have limited liquidity, and face substantial borrowing constraints. For these producers, standing forest is not only a productive asset but also a liquid asset that can be converted into cash through timber extraction, pasture conversion, and land appreciation (Rosenzweig and Wolpin 1993; Fafchamps, Udry, and Czukas 1998; Pfaff 1999; Pattanayak and Sills 2001; Amacher, Koskela, and Ollikainen 2009; Jayachandran 2013; Wunder et al. 2014; Ferraro and Simorangkir 2020). When drought reduces current income and informal financial markets fail to insure against the shock, clearing forest provides an immediate source of liquidity. Drought may therefore increase deforestation not because agriculture becomes more profitable, but because the private value of liquidating forest temporarily rises for financially

constrained households.

This distinction generates a sharp empirical prediction. If drought affects deforestation primarily through agricultural profitability, producers suffering the largest production losses should respond similarly regardless of their financial capacity. If, instead, drought operates through a liquidity channel, the response should be concentrated among financially vulnerable producers even when commercially oriented producers experience comparable or larger agricultural losses. The Brazilian Legal Amazon provides an unusually suitable setting to distinguish between these competing mechanisms because family and commercial farms coexist within the same institutional and climatic environment while differing systematically in their ability to cope with drought shocks.

That distinction matters for policy design because the two mechanisms imply fundamentally different interventions. Brazilian conservation policy has been built largely around changing the profitability of forest clearing through satellite monitoring, environmental enforcement, supply-chain agreements, restrictions on rural credit, and property-rights institutions (Cisneros, Zhou, and Börner 2015; Gibbs et al. 2016; Assunção et al. 2020; Baragwanath and Bayi 2020; Heilmayr et al. 2020; Assunção and Rocha 2019; Souza-Rodrigues 2019; Assunção, Gandour, and Rocha 2023; Baragwanath and Shinde 2026). Each of these policies either raises the private cost of clearing or lowers its expected return. If households instead clear forest because drought temporarily tightens liquidity constraints, then rural credit, crop insurance, and drought-indexed transfers become instruments of climate policy, while enforcement alone may leave the underlying incentive unchanged. Distinguishing between profitability and liquidity is therefore central not only to understanding why drought affects deforestation but also to identifying which policy instruments are likely to remain effective as drought shocks become more frequent in the Amazon.

We test these competing explanations using a municipality-by-grid panel covering the Brazilian Legal Amazon from 2005 through 2019. We combine annual forest loss from the 30-meter MapBiomas Collection 9 with a satellite-based measure of agricultural drought derived from daily root-zone soil moisture (RZSM) simulated by NASA's Global Land Data Assimilation System (GLDAS-2), following Osman et al. (2024).[1] We merge these data with

1. GLDAS-2 is a NASA and NOAA project that creates high-resolution global land surface data. It uses

environmental enforcement records, municipality-level agricultural and livestock production, rural credit records, irrigation maps, agricultural census data, protected areas, commodity prices, and recently developed atmospheric moisture-flow data. The resulting dataset contains 13,160 municipality-by-grid cells spanning 771 municipalities, yielding approximately 197,400 cell-year observations.[2]

Our identification strategy exploits a key feature of the RZSM data. First of all, agricultural drought is fundamentally a shortage of plant-available water rather than simply low rainfall, making root-zone soil moisture the economically relevant state variable for agricultural production (Wilhite and Glantz 1985). GLDAS-2 simulates soil moisture using meteorological forcing (temperature, precipitation, wind, radiation, etc.) and fixed land-surface characteristics, without incorporating contemporaneous deforestation, irrigation, or groundwater extraction. Consequently, our drought measure is likely to be orthogonal by construction to producers' land-use decisions and adaptation responses. Concerning the small feedback loop between deforestation and rainfall via moisture recycling (e.g., Staal et al. 2020), we find no statistically-detectable within-year feedback from local deforestation to local rainfall once we condition on prior forest stock in our panel regression model, suggesting that reverse causality from deforestation to drought is unlikely an empirical issue.

Our findings consistently support the liquidity-constraint mechanism. Drought increases forest clearing, but the response is concentrated entirely among municipalities where family farming predominates or family farm area shares are above the median at the baseline. The municipalities with higher concentration of commercial farms exhibit no detectable increase in deforestation despite experiencing similar drought shocks. In addition, the response operates almost exclusively along the intensive margin, increasing the amount of clearing where deforestation is already occurring rather than substantially expanding clearing onto previously untouched land. These patterns suggest that drought does not broadly increase the profitability of agricultural expansion. Instead, it induces additional clearing precisely among producers most likely to face binding liquidity constraints.

satellite and ground measurements with land surface models to calculate water and energy states on land. Osman et al. (2024) use its RZSM data to measure flash drought.

2. For heterogeneity analyses, we drop one municipality that is not matched with the 2006 Brazil Agricultural Census data.

The mechanism evidence reinforces this interpretation. Agricultural losses occur across both production systems but translate into forest clearing only among financially vulnerable producers. Rice, the principal rain-fed staple cultivated by family farms, experiences large reductions in planted area, harvested area, production, and revenue during droughts in high-family-farm municipalities where drought-induced deforestation occurs. Moreover, drought does not increase rice prices in these municipalities but in low-family-farm municipalities, further ruling out the profitability mechanism. Commercial producers also experience substantial losses—most notably in cattle holdings—yet do not increase forest clearing. The deforestation response therefore tracks financial vulnerability rather than the magnitude of agricultural losses, distinguishing the liquidity channel from a pure profitability mechanism.

The remaining coping margins tell a consistent story. Rather than expanding adaptation investments during droughts, high-family-farm municipalities where clearing rises experience contractions in rural credit, including PRONAF credit, Brazil's principal rural credit program for family farms, while irrigation does not expand. In high-family-farm municipalities producing both rice and cassava (staple crops), producers tend to harvest more drought-tolerant cassava in response to early-season or pregrowing-season drought that reduces rice production, suggesting that family farms may exhaust relatively inexpensive production adjustments before turning to forest clearing (response to growing-season drought) as a source of liquidity.[3] Together, these results indicate that the institutions capable of smoothing temporary income shocks weaken precisely where drought-induced deforestation is most pronounced.

Several additional findings rule out leading alternative explanations. Environmental enforcement intensifies rather than weakens during droughts: both the number and value of forest-related fines increase, particularly in municipalities where deforestation rises, rejecting the hypothesis that the estimated effects reflect reduced monitoring. Likewise, the timing of the response is inconsistent with a flammability-based explanation (more flammable in the dry or pregrowing season). Pregrowing-season drought either reduces or has no effect on clearing, whereas deforestation rises only in response to growing-season moisture deficits that

3. Cassava is a multi-year crop with flexible harvesting time. We define general pregrowing and growing seasons for major temporary crops in the Legal Amazon.

adversely affect the actual crop production in the field.

Our paper makes three contributions to the literature. First, we provide causal evidence on drought-induced deforestation in the Brazilian Amazon, to the best of our knowledge. Staal et al. (2020) documents a positive association between annual deforestation and cumulative water deficits using within-cell variation over time as we do. We additionally control for important confounders, including municipality-specific exposures to commodity prices, policies, protected areas, and prior forest stock. Pimenta, Reimer, and Novan (2025) also studies drought-induced deforestation with a focus on public versus private land tenure, although they do not control for most of the listed confounders.[4] We focus on family versus commercial farms and find family farms tend to clear forest in response to drought while commercial farms may not even in municipalities with high area shares of public forest.

Second, we provide evidence that distinguishes the liquidity channel from the profitability channel for the drought-induced deforestation in the Brazilian Amazon. While recent work shows that adverse weather can accelerate forest clearing (Hultgren et al. 2025; Tabe-Ojong, Tolani, and Molua 2026; Khan et al. 2025; Hsiao et al. 2026), existing explanations primarily emphasize on inelastic food demand for subsistent crops. We show instead that drought-induced deforestation is concentrated among financially constrained producers, even when commercially oriented producers experience comparable production losses. Jayachandran (2013) and Ferraro and Simorangkir (2020) also find evidence on the liquidity channel for deforestation in Uganda and Indonesia where smallholders dominate.

Third, we contribute to the growing literature on climate adaptation and conservation policy. Our results imply that policies designed to discourage profitable forest conversion; including monitoring, enforcement, and market-based restrictions; may not fully address deforestation arising from temporary liquidity shocks (Assunção et al. 2020; Assunção et al. 2023; Araujo, Costa, and Sant'Anna 2025). Because deforestation may reflect households' efforts to smooth income following drought, expanding access to rural finance, insurance, or drought-indexed transfers can complement traditional conservation policies by improving the resilience of vulnerable agricultural households. Our estimates show that agricultural

4. They assume that public and private land share same shocks within each cell. Their 0.5-degree cells, however, can be larger than a municipality in the Brazilian Amazon.

drought accounts for roughly $135 million in carbon damages per year, suggesting substantial room for such policy interventions.

The remainder of the paper proceeds as follows. Section 2 describes the institutional and environmental setting. Section 3 develops the conceptual framework. Sections 4 and 5 describe the empirical strategy and data. Sections 6 and 7 present the main results and mechanism evidence. Section 8 reports robustness analyses, Section 9 discusses welfare and policy implications, and Section 10 concludes the paper.

2 — Background

Deforestation in the Brazilian Amazon. The Brazilian Legal Amazon spans approximately five million square kilometers across nine states and contains the largest contiguous block of tropical forest on Earth. Between 2000 and 2019 the region lost more than 400,000 square kilometers of forest cover, making it one of the most active deforestation frontiers globally and a major source of land-use carbon emissions (Souza Jr. et al. 2020; MapBiomas Project 2024; Ometto et al. 2023). Deforestation rates are highly volatile over time. After peaking in the early 2000s at over 25,000 km^2 per year, annual forest loss fell sharply following the introduction of the Action Plan for the Prevention and Control of Deforestation in the Legal Amazon (*PPCDAm*) in 2004 and remained comparatively low through the early 2010s, before rising again toward the end of the decade (West and Fearnside 2021). This temporal variation reflects the combined influence of commodity prices, environmental governance, land tenure conditions, and infrastructure expansion (Assunção, Gandour, and Rocha 2015; Souza-Rodrigues 2019; Assunção et al. 2020; Baragwanath and Bayi 2020; Baragwanath and Shinde 2025, 2026). Understanding what drives deforestation in any given year thus requires isolating the contribution of each channel—a challenge that motivates the empirical design of this paper.

Anti-deforestation policies. Brazil has deployed a layered set of policy instruments to reduce forest loss in the Amazon. The 2006 Amazon Soy Moratorium, a voluntary supply-chain agreement, committed major traders not to purchase soy grown on land cleared after

July 2008 within the Amazon Biome, effectively extending zero-deforestation requirements to a large share of the commercial soy sector (Gibbs et al. 2016; Heilmayr et al. 2020; Assunção, Gandour, and Rocha 2015; Assunção et al. 2020). The Priority Municipalities (*Municípios Prioritários*, or MPs) program, introduced in 2008, placed the highest-deforesting municipalities on a public blacklist, triggering enhanced monitoring, credit restrictions, and intensified enforcement by IBAMA, the federal environmental agency (Cisneros, Zhou, and Börner 2015; Assunção and Rocha 2019; Baragwanath and Shinde 2025). The 2012 Forest Code established farm-level requirements for maintaining Legal Reserves of native vegetation and Permanent Protection Areas, while also offering amnesty for historical deforestation. Together these programs substantially altered the incentive landscape for land use in the Amazon, and their implementation dates provide important sources of variation that we control for throughout the analysis. We also account for a major 2008 rural credit concession that restricted subsidized credit access for producers within the Amazon Biome, a policy shock that coincided with the Priority Municipalities program and confounds simple before-after comparisons (Assunção et al. 2020; Souza, Herschmann, and Assunção 2020).

Agricultural structure: family farms and commercial producers. The agricultural economy of the Legal Amazon is characterized by two fundamentally different production systems that face distinct financial and technological constraints. At one end are large commercial operations producing soybeans, cattle, and other export-oriented commodities. These farms account for most agricultural land and are the primary targets of environmental monitoring, supply-chain commitments, and formal credit programs. At the other end are family farms, which dominate numerically across much of the Amazon frontier and specialize in labor-intensive production of staple food crops for domestic markets.[5]

These differences imply fundamentally different margins of adjustment following adverse

5. Under Brazil's Family Farming Law (Lei 11.326/2006), family farms are defined as establishments that rely primarily on household labor, derive most household income from the farm, and operate below a municipality-specific size threshold of four fiscal modules. According to the 2006 Agricultural Census, family farms accounted for approximately 84 percent of all agricultural establishments in Brazil, although they occupied only about one-quarter of agricultural land. They produce a disproportionate share of staple crops, including cassava, rice, maize and beans, which are central to Brazil's domestic food supply. Please refer, Assunção et al. (2020), Souza, Herschmann, and Assunção (2020), Leal, Assunção, and Hoover (2026), Souza and Albuquerque (2023), and Instituto Brasileiro de Geografia e Estatística (IBGE) (2006)

climate shocks. Commercial producers generally have access to formal credit, crop insurance, irrigation, and diversified production portfolios that allow temporary income losses to be absorbed without substantial changes in land use. Family farms, by contrast, depend predominantly on rain-fed agriculture, possess limited liquid assets, and face persistent borrowing constraints.[6] These differences motivate the central hypothesis of this paper. When drought reduces agricultural income, commercially oriented producers can smooth consumption through financial markets, whereas liquidity-constrained family farms may instead rely on assets held on the farm itself. In frontier environments, standing forest constitutes one such asset: it can be converted into immediate income through timber extraction, pasture expansion, or by increasing the marketability of land claims. Consequently, identical climate shocks may generate very different deforestation responses depending not on the magnitude of production losses, but on producers' ability to finance those losses without liquidating forest.

Drought in the Amazon. The Legal Amazon has experienced a marked increase in drought frequency and severity over the past two decades. Three major droughts — in 2005, 2010, and 2015–2016 — caused widespread ecological disruption, elevated fire incidence, and large-scale tree mortality across the basin (Anderson et al. 2018; Otkin et al. 2018; Aragão et al. 2018; Yuan et al. 2023). These events were associated with anomalous sea surface temperatures in the tropical Atlantic and Pacific and are consistent with projected trends toward a drier and more variable Amazon climate under continued greenhouse gas forcing (Otkin et al. 2018; Osman et al. 2024; Yuan et al. 2023). Root zone soil moisture deficits during these events (particularly in 2016) were severe and prolonged, with cumulative annual deficits well above historical norms across large portions of the basin. While the ecological consequences of Amazon drought are extensively documented, the behavioral and economic responses of agricultural producers to drought-induced income stress have received comparatively little attention. This paper focuses on that gap, asking whether and how soil

6. The 2006 Agricultural Census reports substantial disparities in access to productive services between family and commercial farms. Fewer than one-fifth of family establishments reported receiving technical assistance, compared with more than 60 percent of non-family farms. Access to formal rural credit and irrigation infrastructure is similarly concentrated among larger commercial producers (Instituto Brasileiro de Geografia e Estatística (IBGE) 2006)

moisture deficits translate into changes in land-use decisions made by different types of farms.

Drought, income shocks, and land use. In rain-fed agricultural systems, soil moisture deficits reduce crop yields and depress farm revenues, generating income shocks that are difficult to insure against in markets characterized by limited financial development. A substantial literature in development economics documents that households and firms facing negative income shocks adopt coping strategies that prioritize short-term liquidity over long-run efficiency, including asset sales, labor reallocation, and changes in production choices (Chomitz and Gray 1996; De Janvry, Fafchamps, and Sadoulet 1991; Pfaff 1999; Rudel et al. 2005; Amacher, Ollikainen, and Koskela 2009). In agricultural frontier regions, forests represent a particularly flexible asset in this context: clearing can generate immediate returns through timber sales, pasture conversion, or speculative land appreciation, and the returns can be realized quickly relative to the time horizon of formal credit or insurance instruments.

This coping logic implies that drought may increase deforestation not by making agriculture more profitable in expectation, but by temporarily raising the relative attractiveness of timber and land conversion income for producers facing short-term cash constraints. If correct, this mechanism has an important implication for conservation policy: enforcement and market-based tools designed for stable economic environments may be insufficient to prevent deforestation during periods of acute climate stress, when private returns to forest clearing are elevated precisely because normal agricultural income channels are disrupted. The empirical analysis that follows tests this mechanism using variation in drought severity across Legal Amazon municipalities and exploits the sharp structural divide between family farms and commercial producers to identify the economic channel.

3 — Conceptual Framework

This section formalizes the coping-response mechanism and derives the empirical predictions that guide the analysis. Our goal is not to establish a fully specified structural model but a transparent framework that clarifies why drought can raise deforestation, and why the response should be concentrated among financially constrained producers.

Setup. Consider a frontier producer i who at the start of period t holds a stock of standing forest and chooses how much of it to clear, $h_{it} \geq 0$.[7] Clearing yields a net expected monetary payoff $R(h_{it})$ from timber sales, land conversion and appreciation, with $R' > 0$ and $R'' < 0$. Standing forest also carries a continuation value V (future agricultural rents, ecosystem services, and the option value of delaying an irreversible decision), so clearing an extra unit of forest forgoes V. Absent frictions, the producer clears up to the point where $R'(h_{it}) = V$, and a negative agricultural shock that lowers expected farm returns tends to lower $R'(h_{it})$ relative to V, *reducing* clearing. This is the standard opportunity-cost logic, and it predicts that drought should slow deforestation. We acknowledge that farm output prices, especially for staple crops, may rise sharply in response to drought that reduces output supply, so that $R'(h_{it})$ can even increase instead (Hsiao et al. 2026).

Liquidity constraint. The key departure is that the producer must also finance current consumption and production out of period-t resources. Let y_{it} denote realized farm income and suppose the household has a subsistence or working-capital requirement $\bar{c}$ that must be met each period, with limited access to credit or insurance to bridge shortfalls. When income is adequate ($y_{it} \geq \bar{c}$) the constraint does not bind and the producer follows the frictionless rule described above. When a drought drives income below the threshold ($y_{it} < \bar{c}$), the constraint binds, and the shadow value of immediate cash from timber sales per unit of cleared forest, $L'(h_{i,t}|y_{i,t}) > 0$, enters the clearing decision: $R'(h_{it}) + L'(h_{i,t}|y_{i,t}) = V$. A binding liquidity constraint scales up the marginal value of the immediate cash that clearing generates. Drought may therefore operate through two offsetting channels: it lowers $R'(h_{it})$, which discourages clearing; but for liquidity-constrained producers ($y_{it} < \bar{c}$), it also raises $L'(h_{i,t}|y_{i,t})$, which encourages clearing. Which channel dominates is an empirical question, and its answer differs sharply across producer types, as illustrated below.

Heterogeneity and predictions. Commercial producers have access to formal credit, crop insurance, irrigation, and diversified production portfolios; for them $L'(h_{i,t}|y_{i,t}) \approx 0$ even in drought years, so the opportunity-cost channel dominates and drought does not necessarily

7. Depending on the location, she or he may or may not face a margin of undesignated public forest, claimed by the state but not allocated, which is in practice open access.

raise clearing. Family farms are rain-fed, uninsured, and credit-constrained; for them drought may push y_{it} below $\bar{c}$ and thereby raise $L'(h_{i,t}|y_{i,t})$, i.e., the liquidity channel kicks in and will dominate when drought is sufficiently severe, such as megadrought. This yields the framework's central prediction and three auxiliary ones that we take to the data:

(P1) **Concentration.** The deforestation response to drought is positive and concentrated among family-farm dominated municipalities, and absent among commercial-farm dominated municipalities.

(P2) **Vulnerability, not loss size.** The response tracks financial vulnerability rather than the magnitude of production loss: commercial farms can suffer large drought losses without clearing, as long as it can smooth the shock.

(P3) **Coping margins.** Where the response is largest, the margins that would otherwise absorb the shock—agricultural output, rural credit, irrigation—are impaired rather than expanded.

(P4) **Real clearing, not measurement.** Because clearing is a behavioral response and is measured from satellite imagery, drought-induced deforestation should be accompanied by more enforcement activity, ruling out a monitoring-failure explanation.

The remainder of the paper tests P1–P4 using data. The sharp structural divide between family and commercial farms in the Brazilian Amazon, as illustrated in Section 2, provides the suitable context for these tests. Before jumping into the data section, we outline our empirical framework in the next section.

4 — Empirical Framework

We use the following dynamic panel regression model to estimate the effects of agricultural drought at the municipality-RZSM (root zone soil moisture) grid or municipality level, depending on the outcomes of interest:

$$Y_{j,t} = \alpha Y_{j,t-1} + \beta DI_{j,t} + C'_{j,t}\gamma + FE_j + FE_t + \varepsilon_{j,t},$$

where (1) Y denotes outcomes of interest, including deforestation (municipality-RZSM grid level), deforestation related environmental fines (municipality level), crop and cattle production (municipality level), irrigation and rural credit use (municipality level); (2) DI denotes the RZSM-based drought intensity (see details in the data section); (3) the control vector C includes municipality-RZSM grid-specific or municipality-specific prior forest stock, protected land area shares, municipality-specific exposure to exogenous crop and cattle prices as well as major municipality or regional level deforestation-related regulation policies (see details in the data section); and (4) FE represents individual (municipality-RZSM grid or municipality) and year fixed effects.

As illustrated below, our primary drought intensity is derived from RZSM data at a resolution of 0.25 degree, corresponding to grid cells of approximately 28 km at the equator. Hence, a municipality can be larger than a RZSM grid cell. Given that, we use the area-weighted drought intensity in the regression for municipality-level outcomes. For municipality-RZSM grid-level outcomes, we use our primary drought intensity directly.

In our main specification, we cluster standard errors at the municipality level, allowing for arbitrary serial correlation within each municipality over time. To account for unobserved or unmeasured spatially correlated shocks, we use Conley spatial standard errors with distance cutoffs ranging from 200 to 500 km as a robustness check. Our drought-deforestation results remain robust.

We include the lagged outcome variable to account for the strong persistence in our outcome variables. We choose a dynamic panel regression as a static panel regression will otherwise produce an estimand with a bias that is notably larger than the dynamic panel regression when the outcome variable is highly autocorrelated (Klosin 2024). Moreover, because our long panel spans fifteen years, the traditional Nickell bias associated with a dynamic panel regression tends to be small (Klosin 2024).

Staal et al. (2020) provide model-based evidence of a small feedback from deforestation to drought in the Amazon basin. Using a moisture-tracking algorithm, they show that deforestation reduces forest evapotranspiration and thereby lowers downwind rainfall, increasing dry-season water deficits during 2003-2014.[8] Smith, Baker, and Spracklen (2023) show that

8. Staal et al. (2020) also documents a positive association between annual deforestation and cumulative

cumulative tropical deforestation over a long period reduces precipitation in neighboring grids, with stronger effects at larger spatial scales. Using cumulative forest loss from 2003 to 2017, they find robust rainfall reductions at scales greater than 50 km, with the largest estimated reduction at 200 km. Our deforestation measure is at the resolution of less than or equal to 0.25^0, corresponding to roughly 28 km at the equator. Also, annual deforestation is small relative to long-term cumulative deforestation. Therefore, the feedback from deforestation to drought is less of a concern in our setting. As expected, we find a negligible and statistically insignificant association between our annual deforestation measure and and local rainfall, obtained from TerraClimate (Abatzoglou et al. 2018). See detailed results in Section 8.

5 — Data

We combine datasets of land cover, root-zone soil moisture (RZSM), irrigation map, administrative environmental fines and agricultural statistics as well as rural credit use to construct two panels for the Brazilian Legal Amazon: one at the municipality-RZSM grid level and one at the municipality level. We use the first panel to document the drought-induced deforestation at a high spatial resolution. We use the second panel to examine economic mechanisms behind it, with a focus on family versus commercial farms.

We analyze the time period from 2005 to 2019. The starting year places our sample entirely within the policy regime established by the Action Plan for the Prevention and Control of Deforestation in the Legal Amazon (PPCDAm), launched in 2004 (West and Fearnside 2021). PPCDAm reorganized federal monitoring, enforcement, and land tenure policy in the Amazon, and the blacklisting and credit-restriction programs we control for below were introduced as policy programs under it. Beginning in 2005 therefore holds the broad enforcement architecture fixed, so that our estimates capture responses to drought within a stable policy environment rather than a mix of drought effects and the transition into a new regime.[9] Two data constraints point the same way: municipal agricultural statistics

water deficits using within-cell variation over time. We aim to provide causal evidence on the drought-induced deforestation by controlling for municipality-specific exposures to commodity prices, policies, protected areas, and prior forest stock, as discussed above.

9. Because PPCDAm applied to the entire Legal Amazon at once, it generates no cross-municipality

and environmental enforcement records are sparse before 2005, and our rural credit data end in 2019. Additionally, because the inclusion of the lagged outcome in our dynamic panel regression requires an initial year to construct the first lag, our effective estimation sample spans 2006–2019.

5.1. Agricultural Drought

Agricultural drought occurs when soil moisture is insufficient to meet the water demands of plants (Wilhite and Glantz 1985). We therefore measure drought using root zone soil moisture (RZSM) rather than a precipitation-based index.

We obtain daily RZSM from NASA's Global Land Data Assimilation System, version 2 (GLDAS-2), which combines satellite observations, ground-based meteorological measurements, and the Catchment Land Surface Model (CLSM) to produce physically consistent estimates on a 0.25-degree grid, roughly 28 km at the equator. We use the series compiled by Osman et al. (2024), who use it to measure global flash drought.[10] RZSM is reported in kilograms per square meter, the total water mass held in the root zone soil column (0–100 cm below surface). This is the layer directly relevant for crop water uptake and plant stress.

Two features of the CLSM make RZSM well suited to identifying exogenous water shocks. First, the model uses a static land cover classification. Specifically, soil and vegetation types do not respond to deforestation, our main outcome.[11] Second, the model does not represent irrigation or groundwater pumping. Our drought measure therefore reflects hydroclimatic water stress generated by meteorological forcing and fixed land surface characteristics, not human adaptive responses.

We construct drought intensity in three steps. We standardize daily RZSM within each

variation and is absorbed by year fixed effects. Restricting the sample to the post-launch period is what makes this absorption meaningful.

10. Osman et al. (2024) use the RZSM field derived from the Catchment Land Surface Model (CLSM) version 2.5. This dataset provides a comprehensive and continuous daily record of RZSM, offering an extensive temporal scope for analyzing environmental and climatic soil moisture trends from 1980 to 2021. They select GLDAS-2 for its enhanced hydrological representation, incorporating GRACE data assimilation to uniquely capture the complex interactions between soil moisture and groundwater for a more comprehensive understanding of the water balance. We use historical RZSM distributions from this dataset to estimate grid-specific drought thresholds based on the bottom 10th percentiles.

11. Base soil and vegetation types are fixed throughout the sample period, although other land surface model parameters may capture historically repeated seasonal cycles within a year.

grid cell using its full historical record (from 1980 to 2021). We then compute daily deficits relative to the cell-specific 10th percentile. We finally sum daily deficits at the grid-cell level within a year or a crop season (details below). Larger values indicate more severe or more prolonged soil moisture stress.

We use drought intensity in two panels: one at the municipality-RZSM grid level, one at the municipality level. For the latter, we use area weights to aggregate raw RZSM-grid level drought intensities within each municipality. The mean annual deficit at the municipality-RZSM grid level is about 3.0 units, but the distribution has a long right tail reaching 548 units in the most extreme cell-years, and 15 percent of cell-years fall below the drought threshold on at least one day.[12] Municipality-level annual RZSM deficits have similar statistics, as reported in Table A1. Figure D1 shows mean annual RZSM deficits across grids, mostly concentrated in the central and northern parts of the Legal Amazon.

We use annual RZSM deficits in our main drought-deforestation and drought-environmental fines regressions. When it comes to the production effects for major temporary crops, such as rice and soybean, we separate drought by crop season to study agricultural mechanisms. Following Rattis et al. (2021), we define a general growing season in the Legal Amazon from September of $t-1$ to June of year t, and a general pregrowing season from July of year $t-1$ to August of year $t-1$.[13] Growing season deficits affect crop development from germination through maturation, while pre-growing season deficits precede the planting window and may shape planting decisions. We also use these seasons to explore certain adaptation margins, such as crop substitution, irrigation and rural credit use.

We acknowledge that drought intensity is highly skewed, with the maximum exceeding at least 80 times its mean, as shown in Table A1. We check the sensitivity of our results to the associated megadrought, specifically the 2015–16 megadrought, as reported in Section 8. We also acknowledge that our drought intensity measure may be subject to measurement errors for two reasons. First, the underlying RZSM data are simulated rather than directly observed so that their levels may differ from each other. Second, whether daily RZSM falls

12. This drought percentage is higher than 10%, as the weather gets hotter and drier in more recent years.

13. USDA uses the same growing season for soybean, mostly grown in the Legal Amazon. Assunção, Gandour, and Rocha (2015) use a similar-but-slightly-shorter growing season, from August of year $t-1$ to May of year t.

below its historical 10th percentile is itself probabilistic so that our drought intensity measure may be noisy. To address the first source of measurement error, we construct an alternative drought intensity measure based on the number of days during which daily RZSM falls below its historical 10th percentile. To address the second source, we redefine our drought intensity over only drought spells lasting at least 7 consecutive days, thereby reducing the influence of short-lived threshold crossings that are more likely to reflect measurement error than persistent drought conditions. Our drought-deforestation results remain robust under these alternative measures, as reported in Section 8.

Why Not SPEI?

The Standardized Precipitation Evapotranspiration Index (SPEI) is the most widely used drought index in economics (Vicente-Serrano, Beguería, and López-Moreno 2010; Reig-Gracia and Latorre Garcés 2023). We prefer RZSM for two reasons. First, RZSM measures the stock of soil water available to plants, whereas SPEI measures a flow of atmospheric water balance. Crops draw on accumulated soil moisture, not on contemporaneous rainfall alone. Second, GLDAS computes actual evapotranspiration from the modeled water balance, whereas SPEI uses potential evapotranspiration. Actual evapotranspiration is closer to the water stress vegetation experiences. For example, during rapid-onset or prolonged droughts, atmospheric demand (PET) increases sharply, leading to extremely low SPEI as the index assumes maximum atmospheric water loss. In reality, however, stressed plants may close their stomata to conserve their water supply, which can significantly reduce actual evapotranspiration (AET). Because plants transpire less, they withdraw less water from the root zone—a biological self-preservation mechanism that slows the rate of RZSM depletion. GLDAS-modeled RZSM explicitly accounts for this constrained AET, making it a more realistic proxy for the actual hydro-agricultural stress experienced by vegetation than SPEI.

Deforestation and Drought

Beyond the interaction of bio-geophysical factors and human-induced land cover changes over time, the spatial-spillovers of forest degradation caused by such effects is equally well-studied in the policy domain. Acknowledging wind patterns as the medium for moisture transport

in the region, Araujo et al. (2023) estimate spatial amplification of degradation caused by deforestation itself in the Amazon. The forests in Brazil not only regulate regional moisture recycling that is an important source of precipitation to its own agricultural areas but also act as the trans-boundary source of precipitation to the downwind agricultural areas in neighboring regions (Leite-Filho et al. 2021; Pranindita et al. 2025).

As evapotranspiration from the tropical forests contributes to around 41% of basin mean precipitation over the Amazon (Baker and Spracklen 2022), the forest-rainfall feedback is important to consider especially at larger spatial scales (200km) where the negative response of precipitation to deforestation is more prominent (Smith, Baker, and Spracklen 2023). At smaller geographical scales, the evidence about the effect of biome-wide deforestation on forest moisture recycling and irreversible biome transition remains uncertain (Leite-Filho et al. 2021). In fact, the strength of the drought-deforestation feedback, which is a function of cumulative historical forest loss (i.e., deforestation minus reforestation), is rather weak at the 0.5° resolution (Staal et al. 2020). Consistent with this, Section 8 shows that annual local forest loss has a limited and statistically insignificant association with local rainfall from TerraClimate (Abatzoglou et al. 2018)

5.2. Deforestation

Our primary outcome is annual forest area converted to any anthropic land covers, measured against a fixed forest baseline. We construct this deforestation measure from MapBiomas Collection 10 annual land use and land cover data at 30-meter resolution. MapBiomas combines Landsat imagery, machine learning classification, and expert validation, and covers Brazil from 1985 onward. We access and process the data through Google Earth Engine.

We identify rainforest pixels in the base year 2000 and measure annual area of anthropic-use land converted from that fixed forest stock over 2005–2019, aggregating to the municipality-RZSM grid-year level.[14] We use land cover code 3 (forest formation) to identify rainforest and use another 17 land cover codes to identify anthropic-use land, including farming and non

14. Our drought-induced deforestation results are robust to alternative base years of 1985, 1990, and 1995. Results are available upon request.

vegetated area.[15] Measuring forest loss against a fixed baseline rather than as a year-to-year net change avoids offsetting clearing against subsequent regrowth, so the outcome reflects conversion of the forest standing in 2000 rather than net cover change. Figure D2 shows mean annual deforestation across grids in the Legal Amazon. Most places have some levels of deforestation, although deforestation is more concentrated in the southern and eastern parts of the Legal Amazon.

As shown in Table A1, mean annual deforestation is 0.91 km^2 per cell-year, with a maximum of 85 km^2; 47 percent of cell-years record no deforestation. We therefore apply the inverse hyperbolic sine transformation, which accommodates zeros while approximating the logarithm at large values. We also report regression results using an indicator for any deforestation in a given year as the outcome variable.

5.3. Environmental Enforcement

Increases in measured deforestation during droughts could reflect changes in clearing behavior or changes in monitoring. To distinguish these two, we use administrative records on forest-related environmental fines issued by the Brazilian Institute of Environment and Renewable Natural Resources (IBAMA). The data report the annual number and monetary value of infractions for illegal deforestation, forest degradation, and unauthorized clearing at the municipality level. We deflate fine amounts using the National Consumer Price Index (IPCA, general) downloaded from IpeaData. As shown in Table A1, the mean municipality-year records 7.2 fines worth 2.0 million in 2005 R$, and roughly half of municipality-years record no fine at all, so both measures enter in inverse hyperbolic sine form. We also report regression results using an indicator for any fine in a given year as the outcome variable.

These records are administrative rather than remotely sensed and so provide an independent measure of government activity. As reported below, we find an increase in fines during drought years. This is inconsistent with a monitoring-failure interpretation and supports the view that observed increases in clearing reflect genuine changes in deforestation behavior.

15. These land cover codes are 9, 15, 20, 21, 24, 25, 30, 31, 35, 39, 40, 41, 46, 47, 48, 62, and 75. You may find the official land cover legend here: https://brasil.mapbiomas.org/wp-content/uploads/sites/4/2025/08/Legenda-Colecao-10-Legend-Code.pdf.

5.4. Who Farms Each Municipality?

Our theoretical framework in Section 3 predicts that family farms are more likely to respond to agricultural drought by deforesting because they are more likely to face liquidity constraints, whereas resource-rich commercial farms are less likely to do so. Hence, the prevalence of family farming in a municipality matters for drought-induced deforestation. We measure it using the 2006 Brazilian Agricultural Census, the most comprehensive source of farm structure available for our period (Instituto Brasileiro de Geografia e Estatística (IBGE) 2006). Brazilian law (Lei 11.326/2006) defines family farms by farm size, reliance on family labor, and household dependence on agricultural income.

For each municipality we calculate the share of agricultural area operated by family farms. As shown in Table A1, the area share averages 33 percent and ranges from less than 1 percent to 97 percent across the 770 Legal Amazon municipalities present in the 2006 Agricultural Census. We classify municipalities with family-farm shares above the Legal Amazon median (29.6 percent) as high-family-farm municipalities and the remainder as low-family-farm municipalities. This classification is fixed across all years. Because it is measured at the base year of our effective sample period (2006-2019), it captures pre-existing differences in agricultural organization rather than endogenous responses to the droughts we study. Figure D3 shows the 2006 family farm distribution by municipality with high family farm area shares in the northern and eastern parts of the Legal Amazon.

5.5. Agricultural Production

We measure crop production using the *Pesquisa Agrícola Municipal* (PAM), an annual survey conducted by the Brazilian Institute of Geography and Statistics (IBGE) (Instituto Brasileiro de Geografia e Estatística (IBGE) 2023a). PAM reports planted area, harvested area, output quantity, and output revenue at the municipality level. We divide output revenue by output quantity to recover average price. We deflate revenues and prices in 2005 R$ using the Broad Producer Price Index (IPA, general) downloaded from IpeaData.

We focus on four temporary crops that span the range of production systems in the Legal Amazon. Soybean and sugarcane are grown predominantly at commercial scale. Rice

and cassava are the primary staple crops of smallholder and family farm households.[16] We restrict each crop sample to municipalities reporting positive production throughout the period, which yields 167 municipalities for soybean, 178 for sugarcane, 496 for rice, and 689 for cassava.

Our data show that, on average across municipality-years, high-family-farm municipalities devote 1.6 times more land to cassava cultivation than low-family-farm municipalities, although low-family-farm municipalities devote about 50% more land to rice cultivation. When it comes to soybean and sugarcane, on average low-family-farm municipalities allocate about 7 times as much land as high-family-farm municipalities at the municipality-year level.

Production outcomes enter in logs, and each regression is weighted by the 2005 municipality-specific production outcome in levels. For average price, we use the 2005 municipality-specific output quantity as weights in the regression. See summary stats of these outcomes in Table A2.

Drought exposure is defined to match each crop's production cycle. Soybean and rice are annual crops, so we use separate pre-growing and growing season deficits. Cassava and sugarcane are continuous multi-year crops, so we use the year-round deficit. A separate sample of 464 municipalities that jointly cultivate rice and cassava lets us test for substitution toward the more drought-tolerant crop (cassava).

We also study the impact of drought on cattle production as it allows us to test whether drought-induced income shocks extend beyond crops, possibly through forage deterioration. We measure livestock activities using the *Pesquisa Pecuária Municipal* (PPM), also compiled annually by IBGE (Instituto Brasileiro de Geografia e Estatística (IBGE) 2023b). We focus on cattle head counts, the dominant livestock activity in the region, restricting to the 767 municipalities with consistently positive reported stocks. See the summary statistics in Table A3. Head counts enter in logs and regressions are weighted by the 2005 municipality-specific head counts in levels.

On average, low-family-farm municipalities raise nearly twice as much cattle as high-

16. We exclude first and second corns, another major crops in the Legal Amazon, because growing seasons differ between first and second corns, making a common drought exposure window difficult to define. Moreover, states dramatically change corn portfolios over time. Without annual production data by corn types, is it empirically infeasible to conduct meaningful regression analyses.

family-farm municipalities. However, the average cattle intensity (head count divided by total farm area, not reported in the summary table) is comparable across the two groups. Our regression results show below, agricultural drought tends to cause greater cattle losses in high-family-farm municipalities.

5.6. Rural Credit

Liquidity constrained producers may borrow to smooth income losses (Jayachandran 2013). Therefore, farmers may use rural credit to cope with agricultural drought. However, family farms may not be able to borrow money from formal credit institutions as they may not have sufficient collateral or may be unwilling to take the risk of losing collateral (Carter 1988; Boucher, Carter, and Guirkinger 2008). Thus, whether agricultural drought will lead to more or less rural credit use is an empirical question.

In Brazil, rural credit is the primary policy instrument for financing agricultural production. Annual planned rural credit is roughly 30 percent of total agricultural production value, and increases in credit supply raise production, productivity, and native forest cover, particularly in the Amazon Biome (Assunção et al. 2020).

We obtain credit data from two sources. For 2005–2011 we use the municipality-level series assembled by Assunção et al. (2020), which reports credit by crop and by cattle. For 2014–2019 we use loan-level records from the Central Bank of Brazil's Rural Credit and Proagro Operations System (SICOR) (Central Bank of Brazil (BCB) 2023), which identify both the purpose and the credit program of each operation. We aggregate loans to the municipality-year level using the agricultural-year definition of Assunção et al. (2020), August of year $t-1$ through July of year t, which closely matches our crop-year window. Because that window requires data from the preceding August, the SICOR series yields complete measures only from 2014, and 2012 and 2013 are unavailable in both sources. Credit regressions therefore omit those years.

We construct total credit amounts and contract counts for crop and cattle production across the full credit sample, and separately identify two programs of particular interest within the SICOR period. PRONAF (*Programa Nacional de Fortalecimento da Agricultura Familiar*) is the principal federal credit program for family farms, which matters because we

find drought effects concentrated in high-family-farm municipalities. Proirriga (*Programa de Financiamento à Agricultura Irrigada e ao Cultivo Protegido*) finances irrigation and protected cultivation, and is used sparsely: mean use rate is about 3%, the mean contract count is close to 0.06, and mean annual disbursement is roughly 32,000 per municipality in 2005 R$, as shown in Table A3. In the credit-use regressions, we therefore use an indicator for Proirriga use at the municipality-year level. We deflate all other credit amounts using the National Consumer Price Index (IPCA, general) downloaded from IpeaData. All credit variables (amounts and counts, except for Proirriga) enter in inverse hyperbolic sine form to accommodate a small fraction of municipality-years with no operations.

Although Proirriga credit is uncommon across municipalities, crop and cattle credit, as well as PRONAF credit, are widely observed (over 90% of municipality-years, regardless of credit lines and family farm prevalence). In terms of credit intensity, measured as the ratio of municipality-year credit amounts to municipality-specific agricultural area in 2006 (not reported in the table), high-family-farm municipalities have higher PRONAF credit intensity, as expected, whereas low-family-farm municipalities have higher crop production credit intensity. However, cattle production credit intensity is comparable across the two groups of municipalities.

5.7. Irrigation

Irrigation is another important channel of drought adaptation in the literature (Taylor 2026). We measure irrigation using the MapBiomas Irrigation Collection, which provides annual 30-meter maps of irrigated agriculture across Brazil from 1985 onward. The dataset is derived from Landsat imagery using machine learning classification and expert validation, and each irrigated pixel carries a code linking it to the underlying land use class in the MapBiomas Land Use and Land Cover Collection, so we observe the crop associated with each irrigated pixel. We aggregate irrigated pixels to the municipality-year level and divide by total agricultural establishment area from the 2006 Agricultural Census, giving an irrigated area share that is comparable across municipalities of very different size. We construct this share for soybean, rice, sugarcane, all other temporary crops. MapBiomas reports no irrigation for cassava, consistent with cassava being rain-fed throughout the Legal Amazon. These measures let

us test whether producers adopt irrigation in response to drought and whether irrigation attenuates drought effects on production and clearing.

As shown in Table A3, irrigation is limited in absolute terms: the mean share across all temporary crops is 0.09 percent of farm area at the municipality-year level. For each crop, the mean irrigation area share among low-family-farm municipalities is at least 3 times higher than that among high-family-farm municipalities. Moreover, the mean irrigation area (not reported in the table) among low-family-farm municipalities is at least 6 times larger than that in high-family-farm municipalities, regardless of crop types. Hence, irrigation is highly concentrated in commercial-farm-dominated municipalities.

5.8. Prices, Policies, and Forest Stock Controls

Drought may coincide with changes in agricultural profitability. Following Assunção, Gandour, and Rocha (2015) and Assunção et al. (2020), we control for exogenous commodity price variation. Price series come from the Paraná State Secretariat of Agriculture and cover beef cattle, soybean, cassava, rice, corn, and sugarcane. Because these prices are set outside local markets, they reflect demand-driven variation rather than local supply responses. We first deflate them to 2005 prices using the Broad Producer Price Index (IPA, general) downloaded from IpeaData. Then, we weight them by each municipality's crop composition in a fixed base period of 2004–2005, and construct separate indices for each semester of the agricultural year, entered contemporaneously and with one lag. The crop index summarizes crop-specific prices by their first principal component; the cattle index enters in levels. Regressions for crop-specific production outcomes use the corresponding crop prices directly.

We additionally control for the major conservation policies in force during our period, most of which were introduced as the policy package of PPCDAm: blacklisting under the Priority Municipalities (*Municípios Prioritários*) program, which covers 3.5 percent of municipality-years; participation in the Green Municipalities (*Município Verde*) program, at 8.7 percent; and the 2008 rural credit restriction and soy moratorium within the Amazon Biome, which applies to 52 percent. For the first two policy controls, we obtain the associated data from Shinde et al. (2025) and Baragwanath and Shinde (2025). We construct the dummy variable for the credit concession and soy moratorium policies within the Amazon Biome using their

common geometric boundary and targeted date (July, 2008).

Unlike PPCDAm itself, these policies apply to some municipalities and not others and are therefore separately identified from year fixed effects. The first two enter with a one-year lag to avoid simultaneity with current deforestation. We also control for the lagged municipality share of protected land, which averages 23 percent, using the World Database on Protected Areas.

Every specification additionally controls for lagged forest stock, which bounds the potential amount of current deforestation. As shown in Table A4, about 98% of our grid-year observations have positive forest stock, with this share rising to 100% at the municipality-year level. We therefore include lagged forest stock in inverse hyperbolic sine form in the grid-by-year regressions and in logarithmic form in the municipality-by-year regressions.

6 — Main Results

6.1. Agricultural Drought and Deforestation

Table B1 reports the average and heterogeneous effects of year-round root zone soil moisture (RZSM) deficits on deforestation across Legal Amazon municipalities from 2006 to 2019. For ease of interpretation, we present the estimated heterogeneous effects for each municipality group (high- and low-family-farm municipalities) rather than the underlying regression coefficients throughout the paper. As outlined in Section 4, both specifications include municipality-by-RZSM-grid fixed effects, year fixed effects, the lagged deforestation, the lagged forest stock, municipality-specific exposure to exogenous crop and cattle prices, protected land area shares, and major deforestation policy programs. We cluster standard errors at the municipality level and our results are robust to Conely spatially-clustered standard errors, as shown in Table B2. The model fit is high overall: the adjusted R^2 of 0.87 reflects the strong predictive role of the lagged outcome and fixed effects, and the within-R^2 of 0.25 indicates a meaningful explanatory power for the within-municipality variation.

On average, we find a positive but statistically insignificant effect of RZSM deficits on deforestation. We find similar results for the average effect at the extensive margin, as

shown in Table B4. However, this muted average effect of drought on deforestation may mask substantial heterogeneity across municipalities with different farm compositions. As we discussed in Section 3, family farms, which typically face tighter liquidity constraints and have limited access to credit and insurance, may be more likely to clear forest as a source of immediate income during agricultural droughts. For heterogeneous effects, we separate municipalities by the prevalence of family farming, defined as whether a municipality's family farm area share lies above or below the 2006 Agricultural Census median (about 30%). Our results remain robust when we directly use the continuous family farm area share in the regressions, as reported in Section 8.

As shown in Table B1, in high-family-farm municipalities, larger year-round RZSM deficits lead to a positive and statistically significant increase in deforestation (p-value $<$ 0.05). Because our regression model is an asinh–linear specification, we follow Bellemare and Wichman (2020) and compute the semi-elasticity implied by the estimated coefficient (0.00022), evaluated at the sample mean of untransformed deforestation among high-family-farm municipalities (0.91).[17] Our calculation indicates a semi-elasticity of 0.00033: a one-unit increase in RZSM deficits leads to 0.033% increase in deforestation.[18]

In contrast, among low-family-farm municipalities, the estimated effect is negative and statistically indistinguishable from zero. The difference between the two groups is statistically significant (p-value $<$ 0.01), indicating that the deforestation response to agricultural drought varies with municipalities' farm composition. The difference in deforestation effects between the two groups is also statistically significant at the extensive margin, as reported in Table B4, although the two group-specific effects are not statistically significant.[19] In sum, the deforestation response to agricultural drought is concentrated in high-family-farm municipalities, with no detectable response among low-family-farm municipalities.

17. Specifically, the semi-elasticity is defined as $\hat{\beta} \times \frac{\sqrt{\bar{y}^2+1}}{\bar{y}}$, where $\hat{\beta}$ is the regression coefficient and $\bar{y}$ is mean untransformed deforestation in the sample. The corresponding elasticity evaluated at sample means is $\hat{\beta} \times \bar{x} \times \frac{\sqrt{\bar{y}^2+1}}{\bar{y}}$, where $\bar{x}$ is mean RZSM deficits in the sample; this equals approximately 0.0012.

18. Chen and Roth (2024) show that if the treatment affects the extensive margin, one can obtain a treatment effect of any magnitude simply by rescaling the units of the outcome variable before taking the inverse hyperbolic sine transformation. This issue is less of a concern in our sample as we do not find significant deforestation effects at the extensive margins. See results below.

19. If anything, larger RZSM deficits tend to decrease deforestation incidence in low-family-farm municipalities (p-value $<$ 0.10).

We examine heterogeneity further by crop season. To align with the calendar year of our deforestation measure, we calculate growing-season RZSM deficits for within-year months excluding July and August and pregrowing-season RZSM deficits for the rest of months within each year.[20] Table B5 shows that larger growing-season RZSM deficits significantly increase (year-round) deforestation among high-family-farm municipalities (p-value < 0.05), with negative-yet-insignificant deforestation response in low-family-farm municipalities. In contrast, larger pregrowing-season RZSM deficits reduce deforestation in both municipality groups, with estimates significant at the 10% level. Table B6 shows a similar pattern for the seasonal deforestation effects at the extensive margins, although the deforestation responses to growing-season RZSM deficits become statistically insignificant. All the heterogeneity results above point to the center role of family farms in the drought-induced deforestation.

We hypothesize that forest clearing functions as a coping response by family farms to drought-induced agricultural income shocks. Family farms mostly rely on rain-fed production systems and are more likely to face binding liquidity constraints, making them more vulnerable to agricultural drought. On the other hand, larger commercial farms may draw on assets or rural credit to cope with drought that reduces their farm incomes. In Section 7, we therefore examine the impacts of agricultural drought on crop and cattle production, irrigation investments, and rural credit use; the results support our proposed hypothesis.

6.2. Agricultural Drought and Environmental Enforcement

A natural concern is that the drought-induced deforestation may reflect a deterioration in monitoring capacity during agricultural drought rather than farmers' behavioral responses to agricultural drought. If drought disrupts the operations of enforcement agencies—through reduced staffing or reallocation of government resources—observed deforestation could increase as the chance of illegal deforestation being caught becomes lower during agricultural drought. Another concern is that whether the legal enforcement tends to be lower for illegal deforestation that is conducted by family farms, which may contribute to the salient drought-induced

20. For crop production, we define a general growing season in the Legal Amazon from September of $t-1$ to June of year t, and a general pregrowing season from July of year $t-1$ to August of year $t-1$, following Rattis et al. (2021).

deforestation among high-family-farm municipalities. We test both hypotheses using data on forest-related environmental fines issued by IBAMA, the Brazilian federal environmental authority.

Table B3 reports the effects of year-round RZSM deficits on fine counts and fine amounts. If drought undermines legal enforcement, we would expect fines to fall during drought years. We find the opposite. Across all Legal Amazon municipalities, larger year-round RZSM deficits significantly raises the count of forest-related fines and their total value (both p-values < 0.01). Enforcement, if anything, increases with higher drought intensity, making it unlikely that our deforestation estimates are driven by monitoring failures.

Consistent with our heterogeneous deforestation results, the increases in fines are statistically significant and larger in magnitude among high-family-farm municipalities, ruling out weaker legal enforcement against illegal deforestation in these areas. Our calculations indicate semi-elasticities of 0.0103 and 0.0045 for fine amounts and counts in high-family-farm municipalities, respectively, evaluated at their sample means of 1033 and 5.8. That is, a one-unit increase in year-round RZSM deficits leads to 1.03% and 0.45% increases in fine amounts and counts, respectively. In low-family-farm municipalities, both estimates are much smaller and statistically insignificant. Fines thus increase where deforestation increases—a pattern that contradicts the monitoring-failure story.

As shown in Table B4, the extensive-margin fine effect largely mirrors the intensive-margin deforestation effect described above. On average, larger year-round RZSM deficits significantly raise the probability of any forest-related fine (p-value < 0.05). The effect is again larger in magnitude and statistically significant (p-value < 0.05) among high-family-farm municipalities.

In line with the seasonal pattern of deforestation responses, larger growing-season RZSM deficits lead to more (year-round) fines and the positive fine responses are more pronounced (both in magnitude and statistical significance) among high-family-farm municipalities. We find these seasonal effects at both the intensive and extensive margins, as shown in Table B5 and Table B6. Larger pregrowing-season RZSM deficits, however, significantly reduce fines in high-family-farm municipalities, consistent with the significant negative effect on deforestation incidence in these municipalities.

In sum, both fine counts and amounts as well as fine incidence rates increase in municipalities where and when forest clearing increases—the opposite of what a monitoring-failure explanation would predict. Putting things together, Tables B1 and B3 establish the core findings of our paper: drought causes deforestation concentrated among high-family-farm municipalities, and this deforestation is both detected by and elicits a response from environmental enforcement agencies. The evidence points to an economic behavioral channel rather than a passive institutional-failure channel. We therefore examine economic mechanisms behind the drought-induced deforestation in the next section.

7 — Mechanisms

The results in Section 6 establish that agricultural drought increases deforestation, with this effect concentrated among high-family-farm municipalities, alongside heightened enforcement responses in these same municipalities. This section investigates the agricultural production channel through which drought translates into deforestation pressure. If forest clearing functions as a coping response to agricultural income shocks, drought should reduce agricultural output—particularly in crops that depend on rain-fed production and in municipalities where family farms predominate. Moreover, high-family-farm municipalities should exhibit limited ability to adapt to agricultural drought, for example, through irrigation investments and rural credit use.

7.1. Agricultural Production

We expect that agricultural drought will have adverse impacts on agricultural production, especially for rain-fed crops. We test this prediction by estimating the effect of RZSM deficits on planted areas, harvested areas, output quantities and revenues for each of the four major crops: soybean, sugarcane, cassava, and rice. We focus on these crops, as they dominate agricultural production in the Brazilian Legal Amazon (Instituto Brasileiro de Geografia e Estatística (IBGE) 2023a).[21]

21. We exclude first and second corns, another major crops in the Legal Amazon, as they have different growing seasons and we do not have production data by corn types at the municipality-year level. States also

Given the geographic diversity of the Legal Amazon, we define the growing season broadly—from September of the previous year through June of the current year—for summer crops (soybean and rice), encompassing planting, crop development, and harvesting activities across the region, following Rattis et al. (2021). We also control for pre-growing-season RZSM deficits to account for any early adaptation responses. For multi-year crops (cassava and sugarcane), we use year-round RZSM deficits as the relevant agricultural drought measure.

To facilitate our discussion, we also examine the effects of drought on cattle production, as cattle ranching is another major driver of deforestation (Assunção et al. 2020). As follows, we present results of drought impacts on crop and cattle production. In particular, we use heterogeneous effects by family-farm prevalence to show that family farms are disproportionately vulnerable to agricultural drought, especially in terms of crop production. As shown in Table A2, we restrict the sample to municipalities with positive crop and cattle production throughout the sample period and take the natural logarithm of production levels, allowing us to interpret the estimated coefficients on RZSM deficits (in levels) as semi-elasticities and facilitating comparisons between high- and low-family-farm municipalities.

7.1.1. Staple Crops: Rice and Cassava

Rice and cassava are the most rain-fed-dependent crops in the Legal Amazon and account for a large share of smallholder farm income (Instituto Brasileiro de Geografia e Estatística (IBGE) 2023a). The adverse impacts of agricultural drought on the production of these two crops provide the clearest evidence of a drought-induced agricultural income shock among economically vulnerable producers. We present associated results below.

Rice. Table C1 shows the sharpest pattern in the crop production effects. Larger pregrowing-season RZSM deficits significantly decrease planted and harvested areas, output quantities, and revenues in high-family-farm municipalities, with no comparable effect detected in low-family-farm municipalities. These findings suggest that family farms may substantially reduce rice production in anticipation of dry soil conditions that could threaten germination and crop establishment later in the growing season. This response likely reflects their limited

dramatically change corn portfolios over time, making it infeasible to conduct meaningful regression analyses.

capacity to invest in irrigation or otherwise mitigate anticipated water stress during planting. In addition, the significant negative effect on rice prices in high-family-farm municipalities may reflect deterioration in rice quality.

Furthermore, larger growing-season RZSM deficits significantly decrease planted and harvested areas and output quantities in high-family-farm municipalities (with a similarly negative but statistically insignificant effect on revenues), but significantly increase these same outcomes in low-family-farm municipalities. One potential reason is that rice producers in low-family-farm municipalities can expand irrigated production in response to anticipated higher rice prices—a margin unavailable to predominantly rain-fed family farms. This is consistent with the significant positive effect of growing-season RZSM deficits on rice prices in those municipalities. High-family-farm municipalities, however, experience negative-yet-insignificant price effects caused by the growing-season drought.

The significant negative effects of RZSM deficits on rice production for high-family-farm municipalities suggest that rice producers in these municipalities tend to be particularly vulnerable to agricultural drought. This likely reflects family farms' limited access to irrigation and credit. Our irrigation data show that high-family-farm municipalities almost have no irrigated cropland for rice (less than 0.02% on average). In Section 7.2, we also show that high-family-farm municipalities significantly reduce rural credit use for crops in response to more severe agricultural drought.

Cassava. Table C2 reports the effects of year-round RZSM deficits on cassava production. Note that cassava is rain-fed in the Legal Amazon, regardless of farm types. Hence, unlike rice, we do not observe opposite production effects between the two municipality groups for cassava. Higher drought intensity reduces planted and harvested area, output quantities, and revenues in both municipality groups (not statistically significant), although there is a positive price effect (statistically significant for low-family-farm municipalities). However, high-family-farm municipalities tend to experience larger negative production effects and smaller positive price effects. Thus, family farms appear more vulnerable to agricultural drought even for cassava production. Nevertheless, agricultural drought has less pronounced negative effects on cassava production than on rice production in high-family-farm municipalities. This difference may

explain why producers in these municipalities increase cassava harvest to compensate for rice output losses caused by pregrowing-season agricultural drought, as reported in Section 7.2.

7.1.2. Cash Crops: Soybean and Sugarcane

Soybean and sugarcane are the dominant commercial crops in the Brazilian agricultural frontier, typically produced at scale with substantial capital inputs. Our irrigation data show that low-family-farm municipalities have much more irrigated cropland for soybean than high-family-farm municipalities (5.26% versus 0.34% on average). However, less than 0.15% of sugarcane cropland is irrigated in both municipality groups, although low-family-farm municipalities have relatively higher area shares of irrigated sugarcane cropland. These features help us understand the associated drought-production effects below.

Soybean. Table C3 reveals a pattern that is broadly similar to, but differs slightly from, the estimated effects on rice production in high-family-farm municipalities. In these municipalities, larger pregrowing- and growing-season RZSM deficits significantly decrease planted and harvested areas and output quantities. However, revenues decline significantly only in response to growing-season drought, which also leads to significantly lower output prices. Pregrowing-season drought leads to a much smaller and statistically insignificant reduction in revenues, potentially reflecting the offsetting effect of higher output prices.

In low-family-farm municipalities, larger pregrowing-season RZSM deficits significantly increase planted and harvested areas, output quantities, and revenues, potentially reflecting anticipated higher output prices, access to irrigation, or the adoption of drought-tolerant soybean varieties. However, larger growing-season RZSM deficits have smaller and statistically insignificant positive effects on planted and harvested areas and output quantities, potentially reflecting a substantially weaker and statistically insignificant positive price response. Nevertheless, revenues increase significantly, consistent with the combined positive, albeit individually insignificant, effects on output quantities and prices.

For high-family-farm municipalities, the positive price effect caused by larger pregrowing-season RZSM deficits almost offsets the negative output quantity effect, making the revenue effect close to null. However, the price effect turns out to be sharply negative for larger growing-

season RZSM deficits, leading to an economically sizable and statistically significant negative revenue effect. These asymmetric price effects make soybean producers in high-family-farm municipalities more vulnerable to the growing-season agricultural drought.

Sugarcane. Table C4 presents the sugarcane results, which differ from all other crops in an important respect. Larger year-round RZSM deficits have almost no effects on planted and harvested areas as well as output quantities in both high- and low-family-farm municipalities. These null effects might reflect features specific to sugarcane's perennial, ratoon-based production system, which differs fundamentally from the replanting cycle of the other crops in our sample. Nevertheless, the significant negative price effect, possibly due to lower sugarcane quality, drives a correspondingly negative revenue effect in low-family-farm municipalities (only statistically significant at 10% level).

7.1.3. Cattle

Previous results show that crop production in low-family-farm municipalities is generally less vulnerable to agricultural drought. The only exception is that sugarcane producers in these municipalities are likely to experience revenue losses due to lower prices. Here, we show that cattle producers in these municipalities are also likely to suffer from agricultural drought, potentially due to forage deterioration. Table C5 shows that larger year-round RZSM deficits significantly reduce cattle head counts in low-family-farm municipalities. Cattle producers in high-family-farm municipalities experience a much smaller and statistically insignificant reduction in head counts. Putting things together, producers in high-family-farm municipalities suffer more from agricultural drought in terms of crop production while producers in low-family-farm municipalities suffer more from agricultural drought in terms of cattle production.

7.2. Adaptation

7.2.1. Crop Diversification and Substitution

Producers may adjust their crop portfolios in response to agricultural drought as crops differ in drought tolerance. For the staple crops considered above, cassava is generally more drought-resilient than rice because it can tolerate relatively long dry spells through deep rooting, reduced leaf area, and flexible growth recovery after rainfall resumes. Rice, in contrast, is more sensitive to water deficits during planting and early growth stages. For instance, rain-fed upland rice requires adequate soil moisture at planting for reliable germination and seedling establishment. Therefore, there are sizable adverse drought effects for (rain-fed) rice production in high-family-farm municipalities, especially during the pregrowing season, but much smaller negative drought effects for cassava production, as we find in the previous section.

For producers who usually cultivate both crops in the Legal Amazon, we may expect that they either plant more drought-tolerant cassava or harvest more of the existing cassava in response to early-season drought conditions. As shown in Table C6, high-family-farm municipalities producing both rice and cassava significantly harvest more cassava in response to larger pregrowing-season RZSM deficits, although they do not significantly plant more cassava. This finding is consistent with resource-constrained producers adjusting along the lower-cost harvesting margin rather than expanding planted area. Low-family-farm municipalities, on the other hand, significantly plant and harvest more cassava in response to the same drought shock, suggesting that better-resourced producers are able to expand cassava production on both margins simultaneously.

7.2.2. Irrigation

Producers may respond to agricultural drought by expanding irrigation. However, whether they can do so contemporaneously remains an empirical question. We obtain irrigation area data for soybean, rice and other temporary crops as well as sugarcane from MapBiomas. For comparison across municipalities, we divide irrigation area by each municipality agricultural establishment area (farm area) to obtain irrigation area share (measured in percentage) and

use it as our outcome variable. Note that soybean and rice are annual crops while sugarcane is a multi-year crop. Considering their difference in the growing cycle, we conduct two sets of regression analyses, one for seasonal drought intensities and one for year-round drought intensities, as we have done in previous sections.

For temporary crops, Table C7 shows that producers may generally decrease irrigation areas in response to more severe growing-season drought, although not statistically significant. This adverse effect tends to more pronounced in high-family-farm municipalities, especially for soybean. Producers in these municipalities also decrease irrigation areas in response to more severe pregrowing-season drought, which effect is statistically significant for soybean. In contrast, producers in low-family-farm municipalities may increase irrigation areas for soybean and rice production to cope with the same seasonal drought shock, although not statistically significant.[22] All these results suggest that family farms may be more vulnerable to agricultural drought as they do not increase but decrease irrigation areas in response to any seasonal drought. Table C7 also shows that larger year-round RZSM deficits may decrease irrigation for sugarcane production in both municipality groups. This effect is not statistically significant though.

Finally, we reestimate the effects of agricultural drought on crop production, additionally controlling for the lagged crop-specific share of irrigated area. The results remain almost unchanged, with no meaningful differences in the magnitudes or statistical significance of the estimated effects across municipality groups.[23] These findings suggest that irrigation plays a limited role in producers' contemporaneous responses to agricultural drought.[24] If anything, producers in high-family-farm municipalities appear to respond to drought by systematically reducing irrigated areas. Nevertheless, we acknowledge that our irrigation measures may be subject to measurement error.

22. They also tend to increase irrigation areas for soybean (but not rice) in response to more severe growing-season drought, although not statistically significant.

23. Results are available upon request.

24. The results remain largely unchanged when examining irrigation responses one year after drought exposure.

7.2.3. Rural Credit

Another commonly-hypothesized risk-coping strategy is rural credit use. In our context, we may expect that producers hit by agricultural drought use more rural credit to smooth consumption or cover production input costs. Theoretically, however, smallholders may not want to take the risk of losing their collateral even when they have access to credit (Boucher, Carter, and Guirkinger 2008); largeholders may have alternative options, such as liquidating asset or wealth (Zimmerman and Carter 2003). Hence, whether producers will use more rural credit or not in response to agricultural drought is an empirical question.

Table C8 shows that on average, producers significantly reduce credit use for crop and cattle production in response to larger year-round RZSM deficits. This negative credit response is concentrated in high-family-farm municipalities, suggesting that family farms may generally respond to drought by using less credit. Producers in these municipalities even reduce the use of PRONAF, a family-farm-specific credit line (p-value < 0.10). In contrast, producers in low-family-farm municipalities do not experience any significant changes in credit use. Regarding the irrigation-investment credit line (Proirriga), producers in both municipality groups seem not to adjust its use in response to larger year-round RZSM deficits.[25]

When we zoom into seasonal patterns, we find that the reduction in credit use for crops mostly responds to growing-season drought, as shown in Table C9. This seasonal negative credit response is also concentrated in high-family-farm municipalities, with low-family-farm municipalities experiencing almost null effects. Producers in high-family-farm municipalities also tend to use less PRONAF credit in response to growing-season drought while they seem to use more of the same credit in response to pregrowing-season drought, although neither effects are statistically significant at the traditional 5% level. Consistent with our year-round findings, producers in both municipality groups generally do not significantly adjust their Proirriga credit use in response to seasonal agricultural drought.[26] The only exception is that

25. We do not find significant changes in Proirriga credit amounts and counts, either. Results are available upon request. In most municipality-years, we do not observe any use of Proirriga; therefore, we use the dummy variable for Proirriga use in our main regressions.

26. Again, this finding also holds true in terms of Proirriga credit amounts and counts. Results are available upon request.

producers in low-family-farm municipalities significantly reduce their adoption of Proirriga credit in response to larger pregrowing-season RZSM deficits.

In sum, our results show that producers in the Legal Amazon generally do not use more credit to cope with agricultural drought. More importantly, producers in high-municipality-farm municipalities significantly reduce credit use for crop and cattle productions, especially in response to growing-season drought. These findings suggest that family farms may face tighter liquidity constraints during growing-season drought, reducing rather than expanding their credit use.

7.3. Synthesis

Our crop production results indicate that agricultural drought tends to hit producers in high-family-farm municipalities more. In particular, they experience significant losses in rice and soybean, while producers in low-family-farm municipalities do not. Although producers in high-family-farm municipalities may offset the rice output loss caused by pregrowing-season drought with more cassava harvest, they are still exposed to significant crop output losses caused by growing-season drought. Moreover, they respond to growing-season drought by significantly decreasing credit use for crops. In contrast, producers in low-family-farm municipalities do not decrease credit use for crops in any season. Taken together, producers in high-family-farm municipalities may partially cope with pregrowing-season drought but not growing-season drought. This asymmetric seasonal coping pattern may explain why only growing-season drought leads to deforestation in high-family-farm municipalities.

Interestingly, producers in low-family-farm municipalities experience significant drought-induced losses in cattle (head count) without responding through deforestation, reflecting their greater capacity to absorb drought shocks. They may cope with agricultural drought through alternative means, potentially their assets or wealth, although they do not significantly use more credit. When it comes to staple crops, they are notably less vulnerable to agricultural drought relative to producers in high-family-farm municipalities. For instance, they significantly plant more cassava in response to more severe pregrowing-season drought whereas producers in high-family-farm municipalities do not. When it comes to growing-season drought, they significantly expand rice production to capitalize on higher prices while

producers in high-family-farm municipalities do not witness such price effect.

Overall, the asymmetric drought responses between these two municipality groups suggest that family farms in the Legal Amazon are likely to be more economically vulnerable to agricultural drought, especially growing-season drought, which may motivate them to clear forest as a coping strategy. Nevertheless, producers in both municipality groups do not notably expand irrigation in response to agricultural drought during any season. If anything, producers in high-family-farm municipalities tend to decrease irrigation for all crops considered in this paper, although not statistically significant at the conventional 5% level. Given that, we expect that drought-induced deforestation will happen repeatedly in high-family-farm municipalities without policy interventions. We discuss policy implications in Section 9.

8 — Robustness Checks

8.1. Feedback from Deforestation to Drought

Recall that we construct our drought measures using daily root-zone soil moisture (RZSM) data from the Global Land Data Assimilation System (GLDAS). As described in the data section, RZSM is generated by the Catchment Land Surface Model (CLSM) driven by meteorological forcing data, including precipitation, temperature, and other weather variables. Although the land surface model relies on static or fixed land surface characteristics, the meteorological forcing data vary from year to year. Hence, an important identification threat to our drought-induced deforestation results is the potential small feedback from local deforestation to local drought via local rainfall (Staal et al. 2020).

Local forest loss can reduce local evapotranspiration, which may in turn reduce local rainfall via moisture recycling (Staal et al. 2020). This forest-rainfall feedback is important to consider especially at larger spatial scales (200km) where the negative response of precipitation to deforestation is more prominent (Smith, Baker, and Spracklen 2023). At smaller geographical scales like ours (0.25 degree), the relationship between forest loss and rainfall remains uncertain (Leite-Filho et al. 2021). We therefore test the relationship between local deforestation and local rainfall using the precipitation data (aggregated at our municipality-RZSM grid level

using area weights) from TerraClimate. Table E1 shows that larger local deforestation has a statistically insignificant association with local rainfall, conditional on prior rainfall and forest stock. This finding remains robust to spatially clustered standard errors. These results suggest that feedback effects from local deforestation to local drought are unlikely to pose a substantial reverse-causality concern in our drought-deforestation regressions.

8.2. Measurement Errors

As explained in Section 5.1, our drought intensity measure—RZSM deficits—is likely subject to measurement errors. We therefore assess the robustness of our main results using two alternative drought intensity measures that are less susceptible to such errors. The first is drought days, defined as the cumulative number of days in a year with daily RZSM below its historical 10th percentile. This alternative measure is to deal with that GLDAS daily RZSM estimates may differ from actual ones. The second is RZSM deficits calculated only over drought spells lasting at least 7 consecutive days. This alternative measure is to deal with that daily RZSM threshold crossings can be random and therefore noisy. As shown in Table E2, our deforestation and environmental fine results remain robust under these alternative drought intensity measures. Higher drought intensities lead to more deforestation and forest-related environmental fines, with the effects being concentrated in or more pronounced among high-family-farm municipalities.

8.3. Continuous Family-farm Area Shares

One natural robustness check is to test whether our documented drought-induced deforestation is increasing in the family-farm area share of a municipality. Table E3 shows that drought-induced deforestation is indeed increasing in the municipality-level family-farm area share at a significance level of 10%. As expected, for municipalities with family-farm area shares close to 100%, the drought-induced forestation effect is statistically significant at 5% level and much larger than our main estimate for high-family-farm municipalities reported in Table B1 ($0.00045 > 0.00022$). Accordingly, we also see much larger forest-related environmental fines for these municipalities, although not statistically significant. Therefore, our drought-induced

deforestation results largely hold when we directly use municipality-specific family-farm area shares for the heterogeneity analysis.

8.4. Sensitivity to the 2015–16 Megadrought

The El Niño megadrought in 2015–16 is the most extreme event in our sample. In 2016, mean RZSM deficits across municipalities reach 19.9, compared with only 1.4 to 2.2 during all other years.[27] A natural concern is whether our drought-induced deforestation results are driven by that single megadrought episode. We probe this directly by re-estimating the model excluding 2015–16; the estimated effect on deforestation is sensitive to this exclusion. Because RZSM deficits are heavily driven by the 2016 spike, excluding this period substantially attenuates the estimated coefficient on RZSM deficits for high-family-farm municipalities (from 0.00022 to -0.00021), rendering it statistically insignificant. Similarly, forest-related environmental fine results also switch signs and lose statistical insignificance. We therefore conclude that, in our sample, the 2015–16 El Niño megadrought is a key driver of drought-induced deforestation.

9 — Discussions

Our main results show that drought-induced deforestation is concentrated in high-family-farm municipalities. We also show that producers in these municipalities experience significant drought-induced crop losses, especially for rain-fed staple crops. In addition, they significantly reduce credit use for crops in response to agricultural drought and appear to reduce irrigation areas for crops. These findings are consistent with our hypothesis (outlined in Section 3) that family farms in the Legal Amazon are likely to be face liquidity constraint under droughts and may consequently clear forests to generate immediate cash through timber sales. Although comprehensive municipality-year data on timber sales are unavailable, we further investigate this potential mechanism by examining heterogeneity in drought-induced deforestation across certain municipality-level socioeconomic characteristics closely associated with family farming.

27. The other two drought years are 2005 and 2010. Mean RZSM deficits in these two years are 2.2 and 2.1, respectively. These numbers are much smaller than that in 2016, as drought in these years are less severe and affected less areas.

These additional findings may also help inform policy implications.

9.1. Rural Household Incomes

Low-income households are more likely to face liquidity constraint when hit by agricultural drought. Hence, it is important to test whether our documented drought-induced deforestation is increasing in the poor-rural-household share of a municipality. We obtain the rural household income distribution data (Tabela 1993) in the 2000 Demographic Census from the Brazilian Institute of Geography and Statistics. This dataset records municipality-level household counts by monthly income bins relative to the minimum monthly wage rate. We classify a rural household with its monthly income less than the minimum wage as a poor household, i.e., for a two-member household, the per-capita monthly income will be less than 1/2 of the minimum wage, the income threshold for the federal registration of poor households.

Column (2) of Table E4 shows that the drought-deforestation response is larger-yet-statistically insignificant among municipalities with rural poor household shares (number of rural poor households/number of rural households) above the median (40%). We also find that the drought-deforestation response is increasing in the poor-rural-household share of a municipality (p-value < 0.10), although not reported here. These findings are consistent with our hypothesis that rural poor households tend to clear forest more than non-poor rural households, in response to agricultural drought.

Our data indicate that there is a moderate positive correlation (0.33) between family farm area shares and rural poor household shares at the municipality level. This small correlation allows us to further explore the interactive roles of family farms and rural poor households in drought-induced deforestation. Column (3) shows that low-family-farm municipalities (fewer family farms) do not deforest more (but less if any) in response to agricultural drought, regardless of rural poor household shares. On the other hand, high-family-farm municipalities respond to drought by deforesting more, although statistically significance differs between municipalities with more and fewer rural poor households.

Interestingly, the deforestation response appears to be the same for high-family-farm municipalities with more and fewer rural poor households (0.00021 versus 0.00023). This finding suggests that family-farm households may clear forest in response to agricultural

drought, regardless of income levels. Our hypothesis is that most family-farm households may be liquidity-constrained.

9.2. Public Forest

As shown in Figure D3, high-family-farm municipalities are mostly located in the northern and eastern parts of the Legal Amazon, whereas low-family-farm municipalities are concentrated in the southern part, the traditional Arc of Deforestation. Although both municipality groups were exposed to severe agricultural drought to some extent, as shown in Figure D1, these two municipality groups may differ in characteristics other than family-farm area shares that could drive drought-induced deforestation. In our regressions, we control for municipality-specific exposure to crop and cattle prices as well as major deforestation regulations. We also examine certain municipality-specific outcomes, such as irrigation and rural credit use, that might capture municipality-specific responses to agricultural drought. However, we may still miss certain factors that could confound our interpretation that family farms drive drought-induced deforestation.

One important factor is forest ownership: relative to private forest, public forest is likely to be deforested more during drought, unless it is protected, as the opportunity cost of deforesting public forest is close to zero for individual farmers (Pimenta, Reimer, and Novan 2025; Baragwanath and Shinde 2026). We obtain the public land data from the CNFP (Cadastro Nacional de Florestas Públicas, 2024 edition, rectified July 2025). This dataset includes public land (mostly forest) that is either designated (i.e., indigenous territories, conservation units, military land, federal agrarian settlements, etc.) or undesignated (not yet allocated to a specific use). Designated public land is generally protected and held by communities or government agencies, whereas undesignated public land has been claimed by the government but not yet assigned any such status. We therefore focus on undesignated public land given our research purpose.

Column (2) of Table E5 shows that municipalities with public land area shares above the median (0.01) appear to deforest more in response to agricultural drought relative to municipalities with public land area shares below the median, although the difference is not statistically significant (not reported here). Column (1) is the same as that reported in

Table B1: municipalities with family farm area shares above the median (more family farms) deforest more in response to agricultural drought while the other municipality group does not, and the difference is statistically significant (p-value < 0.01). Thus, family farms may play a more important role than public land.

Our data indicate that there is a small positive correlation (0.25) between family farm area shares and public land area shares (public land area/total land area) at the municipality level. This limited correlation allows us to further explore the interactive roles of family farms and public land in drought-induced deforestation. Column (3) shows that low-family-farm municipalities (fewer family farms) do not deforest more (but less if any) in response to agricultural drought, regardless of public land area shares. On the other hand, high-family-farm municipalities respond to drought by deforesting more, regardless of public land area shares (0.00015 versus 0.00023), although the estimated effect is only statistically significant for high-family-farm municipalities with public land area shares above the median (0.00023 with p-value < 0.05). Overall, these findings suggest that family farms tend to play a central role in drought-induced deforestation while more public land may strengthen that role as it can lower the private opportunity cost of deforestation.

9.3. Policy Implications

Our deforestation estimates carry a direct policy reading: because drought-induced deforestation is likely to be a coping response to a negative liquidity shock, the associated carbon externality is potentially *avoidable* through instruments that relax the constraint. We make this concrete in three steps, deferring the full accounting to Appendix F.

Avoidable drought-induced deforestation. The heterogeneous drought-deforestation effects in Table B1 show that drought led to a statistically insignificant reduction in deforestation in low-family-farm municipalities but a statistically significant increase in high-family-farm municipalities.[28] The additional drought-induced deforestation in high-family-farm municipalities may therefore be *avoidable* through policies that mitigate the negative liquidity shocks

28. We do not use the estimated average-effect dynamic model to calculate drought-induced deforestation as the average effect is not statistically significant and masks substantial heterogeneity between high- and low-family-farm municipalities.

potentially experienced by family farms during agricultural droughts. For each municipality-RZSM grid cell, we simulate the inverse hyperbolic sine-transformed clearing trajectory under two scenarios: one with the high-family-farm municipality indicator set to zero and the other with it set to one.[29] We then apply the hyperbolic sine transformation and sum the differences in predicted clearing across the two scenarios over 2006–2019.[30] This cumulative difference represents our estimate of the *avoidable* drought-induced deforestation in high-family-farm municipalities. Our calculation shows that drought led to about 331 square kilometers of *avoidable* deforestation relative to the year-2000 forest baseline, or about 2,364 hectares per year. The estimate is robust to alternative forest base years (1985, 1990, 1995, and 2000 all deliver similar totals), reflecting that drought accelerates clearing of older, higher-value forest. Although this is a small share of total Legal-Amazon deforestation during 2006-2019 ($\approx 0.2\%$), it is highly concentrated: in the 2016 megadrought year, *avoidable* drought-induced deforestation reached above 3.7 times the annual average.

Committed carbon and its social cost. Converting hectares to committed emissions using an aboveground-biomass emission factor of approximately 300 tons of CO_2 per hectare (Ometto et al. 2023; IPCC 2019), drought-induced clearing commits on the order of 0.71 million tons of CO_2 per year. Valued at a central social cost of carbon of \$190 per ton (U.S. Environmental Protection Agency 2023), this implies a committed carbon damage of roughly \$134.9 million per year on average, rising to about \$505.2 million in the 2016 megadrought year.[31] Because our emission factor counts aboveground live biomass only, and because our counterfactual holds constant the many other drivers of clearing, these figures should be read as a conservative, indicative lower bound on the climate externality of Amazon drought operating through the agricultural channel.

Policy leverage: rural finance and public-land protection. The central policy implication follows directly from the mechanism we identify rather than the magnitude of

29. See Appendix F.1 for details.

30. In our dynamic model, the difference in forest stock for the initial year 2005 is mechanically zero.

31. The sensitivity of these figures to the biomass and social-cost-of-carbon assumptions can be examined based on alternative benchmark numbers reported in Appendix F.2 and F.3 (Interagency Working Group on Social Cost of Greenhouse Gases 2021; Nordhaus 2017; Barrage and Nordhaus 2024).

the estimated effects. Enforcement and market-based conservation policies—the workhorses of Amazon forest protection—are designed to discourage profitable forest conversion. They are less effective when deforestation is driven by temporary liquidity needs following adverse weather shocks. Our falsification exercise reinforces this interpretation: environmental fines increase during drought years, implying that additional clearing occurs despite, rather than because of, weaker monitoring. Simply intensifying enforcement is therefore unlikely to eliminate the coping motive underlying drought-induced deforestation.

Policies that relax short-run financial constraints may instead target the mechanism directly. Expanding access to rural credit, crop insurance, or drought-indexed transfers can reduce the need for liquidity-constrained households to finance temporary income losses by liquidating forest assets. In Brazil, however, access to these instruments remains highly unequal. Despite PRONAF being the country's principal rural credit program, only a small minority of family farmers receive formal credit, with access particularly limited among smaller producers and in the North, where climate vulnerability is greatest (Souza and Albuquerque 2023). Recent evidence further shows that expanding technical assistance (ATER) and strengthening producer cooperatives substantially increase access to PRONAF, suggesting that financial inclusion depends not only on the availability of subsidized credit but also on complementary rural institutions that lower informational and administrative barriers to borrowing (Leal, Assunção, and Hoover 2026). Rather than viewing rural finance solely as a development or social protection policy, our results suggest that improving access to financial services for liquidity-constrained households can also complement climate mitigation by reducing the incentive to clear forest following temporary income shocks.

Our results further highlight the importance of public-land governance. The public-forest results in Section 8 show that drought-induced deforestation is concentrated in municipalities with both a high prevalence of family farming and above-median shares of undesignated public land. This pattern suggests that weakly protected public forests provide the primary margin of adjustment when financially constrained households respond to climate shocks. Strengthening the protection and formal designation of undesignated public lands can therefore complement rural finance policies by limiting opportunities for coping-driven encroachment onto low-opportunity-cost forest.

More broadly, our findings suggest that financial resilience and forest protection should be viewed as complementary components of climate policy rather than independent objectives. Expanding access to rural finance, insurance, or extension services can reduce the incentive to liquidate forest following adverse weather shocks, while stronger protection of undesignated public lands limits the availability of forest as a coping asset. Together, these policies address both the demand for forest clearing created by household liquidity constraints and the supply of vulnerable public forest available for conversion.

10 — Conclusion

This paper studies whether and how agricultural drought leads to deforestation in the Brazilian Legal Amazon where family and commercial farms coexist. Using high-resolution root-zone soil moisture and land cover data, we find that increases in soil moisture deficits lead to higher deforestation, with no statistically detectable within-year feedback from local deforestation to local moisture inflow or rainfall once we condition on prior forest stock. Importantly, the deforestation response is overwhelmingly concentrated where family farming predominates, although both family-farm-predominant and commercial-farm-predominant municipalities experience significant crop or livestock losses. Furthermore, producers in high-family-farm municipalities respond to more severe agricultural drought by reducing, rather than increasing, crop production, suggesting that their deforestation response is likely a signature of binding liquidity constraints rather than of profitability. Finally, these producers are also severely constrained in terms of adaptation: they neither invest more in irrigation nor use more rural credit, suggesting an urgent need for policy intervention.

These findings carry a distinct policy message. The dominant instruments of Amazon conservation—command-and-control enforcement and market-based incentives—are built for stable economic conditions and do not reach the coping motive that drives clearing when smallholder agricultural income collapses. Because drought-induced clearing is likely a response to a liquidity shock, it is avoidable through instruments that relax the constraint: affordable or accessible rural credit, crop insurance, or drought-indexed transfers. Our carbon accounting indicates that the stakes are material—drought-attributable clearing commits on

more than 0.7 million tons of CO_2 per year, worth about $135 million per year at a central social cost of carbon and far more in megadrought years, suggesting substantial room for such policy interventions. Overall, our results suggest that climate adaptation for smallholders and climate mitigation for the planet, in the Brazilian Amazon, are complements rather than competing claims on the policy budget.

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A — Summary Statistics Tables

Table A1 — Summary Stats of Deforestation, Drought Intensity, and Family Farm Data

	N	Mean	SD	Min	Max
Panel A. Municipality-RZSM grid-by-year data (2005-2019)					
deforestation2000 (km2)[a]	197400	0.9133	3.0042	0	84.9787
deforestation2000_dummy	197400	0.5298	0.4991	0	1
deforestation2000_asinh	197400	0.4000	0.7843	0	5.1356
RZSM_dintensity[b]	197400	2.9662	17.5641	0	548.1848
RZSM_ddays[b]	197400	2.0411	6.9667	0	86
RZSM_drought_dummy	197400	0.1530	0.3599	0	1
Panel B. Municipality-by-year data (2005-2019)					
fine_dummy	11565	0.4952	0.5000	0	1
fine_amount (1000 R$, real)[c]	11565	1962.5104	12900.7619	0	688877.0625
fine_amount_asinh	11565	2.9921	3.5974	0	14.1360
fine_count	11565	7.2472	25.0234	0	841
fine_count_asinh	11565	1.1922	1.5057	0	7.4277
muni_RZSM_dintensity[d]	11565	2.3849	10.6657	0	204.2956
muni_RZSM_ddays[d]	11565	1.6634	5.3052	0	47.4442
muni_drought_dummy	11565	0.2610	0.4392	0	1
Panel C. Family farm data (2006 Agricultural Census)					
familyfarm_area (hectares)	770	33051.31	36112.35	110	247487
farmarea_tot (hectares)	770	150213.2	190151.6	756	1581759
familyfarm_areashare (%)	770	33.2707	21.3532	0.0785	97.0252

Note: [a]We calculate annual deforestation (the anthropic conversion of forest) against the baseline forest in 2000 based on the land cover data of the MapBiomas Collection 10. [b]We define RZSM_dintensity as year-round daily RZSM deficits relative to its 10th percentile based on the standardized cell-specific daily RZSM obtained from Osman et al. (2024). RZSM_ddays is the total number of days with RZSM below its 10th percentile in a year. [c]We obtain forest-related environmental fines issued by the Brazilian Institute of Environment and Renewable Natural Resource. We deflate fine amounts to 2005 prices using the National Consumer Price Index (IPCA, general) downloaded from IpeaData. [d]We use RZSM cell-specific area weights to aggregate raw RZSM-grid level drought intensities, including RZSM_dintensity and RZSM_ddays, within each municipality.

Table A2 — Summary Stats of PAM Crop Production Data

	N	Mean	SD	Min	Max
Panel A. Municipality-by-year crop production data: Soybean (2005-2019)					
area_planted (hectares)	2505	51824.4056	86732.2096	60	635000
area_planted_log	2505	9.6412	1.7521	4.0943	13.3614
area_harvested (hectares)	2505	51753.7242	86581.2852	60	630000
area_harvested_log	2505	9.6396	1.7526	4.0943	13.3535
quantity (metric tons)	2505	158126.6930	272845.8923	162	2232000
quantity_log	2505	10.7048	1.7937	5.0876	14.6184
revenue (1000 R$, real)	2505	75072.4554	129791.1007	86.6370	1112666.2352
revenue_log	2505	9.9578	1.8017	4.4617	13.9223
Panel B. Municipality-by-year crop production data: Sugarcane (2005-2019)					
area_planted (hectares)	2670	1516.6813	5863.1935	1	53861
area_planted_log	2670	3.9095	2.1728	0	10.8942
area_harvested (hectares)	2670	1505.9101	5820.4349	1	53861
area_harvested_log	2670	3.8898	2.1741	0	10.8942
quantity (metric tons)	2670	104659.9273	425359.1117	15	4250761
quantity_log	2670	7.4221	2.4557	2.7081	15.2626
revenue (1000 R$, real)	2670	4321.8338	16529.2075	1.3290	176689.0211
revenue_log	2670	4.7343	2.3412	0.2845	12.0821
Panel C. Municipality-by-year crop production data: Cassava (2005-2019)					
area_planted (hectares)	10335	904.7941	2083.6334	3	45000
area_planted_log	10335	5.6360	1.5542	1.0986	10.7144
area_harvested (hectares)	10335	888.8364	2063.7582	3	45000
area_harvested_log	10335	5.6136	1.5548	1.0986	10.7144
quantity (metric tons)	10335	12373.8988	31966.3296	21	900000
quantity_log	10335	8.1563	1.5884	3.0445	13.7101
revenue (1000 R$, real)	10335	2876.7651	7227.0074	3.5057	151198.0397
revenue_log	10335	6.6669	1.6962	1.2544	11.9263
Panel D. Municipality-by-year crop production data: Rice (2005-2019)					
area_planted (hectares)	7440	1594.8905	3446.3446	1	91229
area_planted_log	7440	6.0941	1.8272	0	11.4211
area_harvested (hectares)	7440	1572.2215	3279.0199	1	53105
area_harvested_log	7440	6.0816	1.8290	0	10.8800
quantity (metric tons)	7440	3841.0032	13207.1700	1	366846
quantity_log	7440	6.5340	2.0549	0	12.8127
revenue (1000 R$, real)	7440	1574.3185	5494.8613	0.4766	172182.2828
revenue_log	7440	5.6411	2.0672	-0.7412	12.0563

Table A3 — Summary Stats of Other Outcomes

	N	Mean	SD	Min	Max
Panel A. Municipality-by-year cattle production data (2005-2019)[a]					
head_count	11505	103593.5	157921.1	10	2282445
head_count_log	11505	10.5203	1.7201	2.3026	14.6408
Panel B. Municipality-by-year rural credit data (2005-2011 & 2014-2019)[b]					
credit_amounts_crop (1000R$, real)	9789	7066.1205	27049.9278	0	551698.6644
credit_amounts_crop_asinh	9789	6.4940	3.0973	0	13.9139
credit_contracts_crop (count)	9789	237.3666	469.4125	0	9071
credit_contracts_crop_asinh	9789	4.7131	2.1367	0	9.8060
credit_amounts_cattle (1000R$, real)	9789	5529.6168	9467.5421	0	274585.8870
credit_amounts_cattle_asinh	9789	7.7037	2.6691	0	13.2162
credit_contracts_cattle (count)	9789	303.4442	478.9730	0	9071
credit_contracts_cattle_asinh	9789	5.3844	1.8705	0	9.8060
amounts_pronaf (1000R$, real, 2014-2019)	4614	1873.3800	3373.6984	0	46120.8711
amounts_pronaf_asinh (2014-2019)	4614	6.7445	2.3677	0	11.4322
contracts_pronaf (count, 2014-2019)	4614	188.5847	260.4328	0	2951
contracts_pronaf_asinh (2014-2019)	4614	4.8065	1.9333	0	8.6830
amounts_proirriga (1000R$, real, 2014-2019)	4614	32.0179	325.1093	0	9730.5137
contracts_proirriga (count, 2014-2019)	4614	0.0557	0.4244	0	10
proirriga_dummy (2014-2019)	4614	0.0290	0.1679	0	1
Panel C. Municipality-by-year irrigation data (2005-2019)[c]					
alltemp_irrigated_areashare (%)	11550	0.0876	0.7495	0	27.3691
soybean_irrigated_areashare (%)	11550	0.0280	0.2291	0	7.7409
rice_irrigated_areashare (%)	11550	0.0340	0.6801	0	27.3491
sugarcane_irrigated_areashare (%)	11550	0.0007	0.0299	0	1.8775
othertemp_irrigated_areashare (%)	11550	0.0249	0.1843	0	5.5353

Note: [a]We obtain cattle head count data from PPM, compiled by IBGE. [b]We obtain rural credit data from Assunção et al. (2020) (2005-2011) and the Central Bank of Brazil's Rural Credit and Proagro Operations System (2014-2019). Assunção et al. (2020) do not provide PRONAF and Proirriga credit use data for 2005-2011. PRONAF is the principal federal credit program for family farms while Proirriga specifically finances investments in irrigation and protected cultivation. We deflate all credit amounts using the National Consumer Price Index (IPCA, general) downloaded from IpeaData. [c]We obtain irrigation data from the MapBiomas Irrigation Collection. We divide annual crop-specific irrigation area by the 2006 municipality-specific agricultural area (Agricultural Census) to obtain irrigation area share series.

Table A4 — Summary Stats of Control Variables

	N	Mean	SD	Min	Max
Panel A. Municipality-RZSM grid-by-year forest stock (2005-2019)[a]					
foreststock (km2)[a]	197400	241.7639	261.4766	0	769.3582
foreststock_dummy	197400	0.9797	0.1410	0	1
foreststock_asinh	197400	4.9595	2.1166	0	7.3387
Panel B. Municipality-by-year policies (2005-2019)[b]					
muni_foreststock (km2)	11565	4131.4737	11495.4134	8.0463	145174.9531
muni_foreststock_log	11565	6.6587	1.7593	2.0852	11.8857
protected_areashare (0–1)	11565	0.2258	0.3092	0	1
green (dummy)	11565	0.0874	0.2825	0	1
blacklist (dummy)	11565	0.0353	0.1845	0	1
creditsoy2008 (dummy)	11565	0.5181	0.4834	0	1
Panel C. Municipality-by-year crop and cattle real prices (2005-2019)[c]					
sem1_rprice_index_crop	11565	-0.0351	1.2593	-0.8363	18.9797
sem1_rprice_index_crop_lag	10794	-0.0269	1.2703	-0.8349	19.0799
sem2_rprice_index_crop	11565	-0.0393	1.2544	-0.8319	18.9989
sem2_rprice_index_crop_lag	10794	-0.0349	1.2579	-0.8323	18.8887
sem1_rprice_index_cattle	11565	40.2700	45.1783	0.0000	345.7506
sem1_rprice_index_cattle_lag	10794	40.0586	45.0035	0.0000	345.7506
sem2_rprice_index_cattle	11565	41.3486	46.1448	0.0000	335.3997
sem2_rprice_index_cattle_lag	10794	41.0357	45.8250	0.0000	335.3997

Note: [a]We calculate annual stock of all current forest based on the land cover data of the MapBiomas Collection 10. [b]We calculate protected area share as the ratio of protected area to the total land area using the protected area data from the World Database on Protected Areas. We obtain data for municipalities enrolled in the Green Municipalities program and blacklisted in the Priority Municipalities program from Shinde et al. (2025) and Baragwanath and Shinde (2025). We construct the dummy variable for the credit concession and soy moratorium policies within the Amazon Biome using their common geometric boundary and targeted date (July, 2008). [c]We construct municipality-specific exposure to exogenous commodity prices outside of the Legal Amazon following Assunção, Gandour, and Rocha (2015) and Assunção et al. (2020). See detailed descriptions about the data construction in Section 5.8.

B — Deforestation Results Tables

Table B1 — Impacts of Year-Round RZSM Deficits on Deforestation[a]

	average effects	heterogeneous effects
all Legal Amazon municipalities	0.00008	
	(0.00008)	
high-family-farm municipalities[b]		0.00022**
		(0.00009)
low-family-farm municipalities		-0.00012
		(0.00010)
difference between the two		0.00034***
		(0.00012)
one lag of the outcome variable	YES	YES
one lag of forest stock (*asinh*)	YES	YES
price controls[c]	YES	YES
policy controls[d]	YES	YES
year fixed effects	YES	YES
municipality-by-RZSM-grid fixed effects[e]	YES	YES
$adj.R^2$	0.87	0.87
within R^2	0.25	0.25
N	184,240	184,030[f]

Note: [a]We construct our deforestation measure based on the annual land cover data obtained from MapBiomas Collection 10. We define deforestation in year t as forest land in year $t-1$ converted to any anthropic use in year t. To deal with forest regrowth, we focus on deforestation relative to the base year of 2000 from 2005 to 2019 in the Legal Amazon. We take *asinh* to deal with zero values (close to 50% of the raw deforestation data). [b]We classify a municipality as a high-family-farm municipality if its baseline family farm area share is above the median. We calculate family farm areas and their shares within each municipality based on the 2006 Brazilian Agriculture Census data. [c]We control for local exposures to exogenous crop and cattle prices following Assunção, Gandour, and Rocha (2015) and Assunção et al. (2020). [d]We control for major municipality, regional, national and international regulation policies, including blacklisting and green municipality programs, the credit concession and soy moratorium inside the Amazon Biome, and national and international protected area recognitions. For the first two, we use their lagged implementation statuses. For protected area, we use the one-year lag of the area share as the control. [e]In total, we have 13,160 municipality-by-RZSM-grid cells and 771 municipalities. [f]One Legal Amazon municipality does not show up in the 2006 Brazilian Agriculture Census data. Standard errors, clustered at the municipality level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B2 — Conley Spatial Standard Errors for The Drought-Induced Deforestation Results

	coefficient	clustered standard errors		
	estimates	municipality	Conley (200 km)	Conley (500 km)
all Legal Amazon municipalities	0.00008	(0.00008)	(0.00008)	(0.00009)
high-family-farm municipalities	0.00022	(0.00009)	(0.00010)	(0.00009)
low-family-farm municipalities	-0.00012	(0.00010)	(0.00011)	(0.00011)

Note: We report three clustered standard errors in parentheses. One is clustered at the municipality level and two are Conley spatial standard errors with 200 km and 500 km cutoffs.

Table B3 — Impacts of Year-Round RZSM Deficits on Forest-Related Environmental Fines[a]

	fine amounts		fine counts	
	average effects	heterogeneous effects	average effects	heterogeneous effects
all Legal Amazon municipalities	0.0082***		0.0034***	
	(0.0031)		(0.0012)	
high-family-farm municipalities		0.0103**		0.0044***
		(0.0040)		(0.0016)
low-family-farm municipalities		0.0044		0.0017
		(0.0043)		(0.0014)
difference between the two		0.0059		0.0028
		(0.0057)		(0.0021)
one lag of the outcome variable	YES	YES	YES	YES
one lag of forest stock (*asinh*)	YES	YES	YES	YES
price controls	YES	YES	YES	YES
policy controls	YES	YES	YES	YES
year fixed effects	YES	YES	YES	YES
municipality fixed effects	YES	YES	YES	YES
$adj.R^2$	0.63	0.63	0.69	0.69
within R^2	0.01	0.01	0.03	0.03
N	10,794	10,780[b]	10,794	10,780

Note: [a]We obtain forest-related environmental fines from the Brazilian Institute of the Environment and Renewable Natural Resources. We take *asinh* to deal with zero values (about 50% of the raw fine data). We deflate fine amounts using the National Consumer Price Index (IPCA, general) downloaded from IpeaData. [b]One Legal Amazon municipality does not show up in the 2006 Brazilian Agriculture Census data. Standard errors, clustered at the municipality level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B4 — Impacts of RZSM Deficits on Deforestation and Forest-related Environmental Fines at the Extensive Margin[a]

	deforestation		environmental fines	
	avg. effects	hetero. effects	avg. effects	hetero. effects
all Legal Amazon municipalities	0.00001		0.00128**	
	(0.00003)		(0.00051)	
high-family-farm municipalities		0.00005		0.00150**
		(0.00003)		(0.00071)
low-family-farm municipalities		-0.00005*		0.00091*
		(0.00003)		(0.00051)
difference between the two		0.00010**		0.00059
		(0.00005)		(0.00084)
one lag of the outcome variable	YES	YES	YES	YES
one lag of forest stock (*asinh*)	YES	YES	YES	YES
price controls	YES	YES	YES	YES
policy controls	YES	YES	YES	YES
year fixed effects	YES	YES	YES	YES
municipality-by-RZSM-grid fixed effects	YES	YES		
municipality fixed effects			YES	YES
$adj.R^2$	0.81	0.81	0.45	0.45
within R^2	0.02	0.02	0.01	0.01
N	184,240	184,030	10,794	10,780

Note: [a]Same regressions as in Table B1 and B3 except that we replace deforestation and fine outcomes with their dummies. Standard errors, clustered at the municipality level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B5 — Impacts of Seasonal RZSM Deficits on Deforestation and Forest-Related Environmental Fines[a]

	deforestation		fine amounts		fine counts	
	avg. effects	hetero. effects	avg. effects	hetero. effects	avg. effects	hetero. effects
Pregrowing-season RZSM deficits						
all Legal Amazon	-0.00073*		0.1467		0.0547	
municipalities	(0.00040)		(0.2258)		(0.1006)	
high-family-farm		-0.00042*		-0.3968***		-0.1873***
municipalities		(0.00022)		(0.1070)		(0.0371)
low-family-farm		-0.00131*		0.3253		0.1340
municipalities		(0.00072)		(0.2162)		(0.0970)
difference between the two		0.00089		-0.7221***		-0.3214***
		(0.00073)		(0.2410)		(0.1040)
Growing-season RZSM deficits						
all Legal Amazon	0.00008		0.0081***		0.0034***	
municipalities	(0.00008)		(0.0031)		(0.0012)	
high-family-farm		0.00023**		0.0106***		0.0045***
municipalities		(0.00009)		(0.0040)		(0.0016)
low-family-farm		-0.00012		0.0042		0.0016
municipalities		(0.00010)		(0.0043)		(0.0014)
difference between the two		0.00034***		0.0063		0.0029
		(0.00012)		(0.0057)		(0.0020)
lagged outcome	YES	YES	YES	YES	YES	YES
controls	YES	YES	YES	YES	YES	YES
fixed effects	YES	YES	YES	YES	YES	YES
$adj.R^2$	0.87	0.87	0.63	0.63	0.69	0.69
within R^2	0.25	0.25	0.01	0.01	0.03	0.03
N	184,240	184,030	10,794	10,780	10,794	10,780

Note: [a]We use the same deforestation and environmental fine measures as in Table B1 and B3. But we replace year-round RZSM deficits with two seasonal RZSM deficits, one for the pregrowing season (July-August) and one for the growing season (all other months in the same year). Regressions include individual and time fixed effects, lagged dependent variable, and various controls as the other tables above. Standard errors, clustered at the municipality level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B6 — Impacts of Seasonal RZSM Deficits on Deforestation and Forest-Related Environmental Fines at the Extensive Margin[a]

	deforestation		environmental fines	
	avg. effects	hetero. effects	avg. effects	hetero. effects
Pregrowing-season RZSM deficits				
all Legal Amazon	-0.00117		-0.00795	
municipalities	(0.00095)		(0.04198)	
high-family-farm		-0.00054***		-0.14409***
municipalities		(0.00015)		(0.01446)
low-family-farm		-0.00202		0.03683
municipalities		(0.00214)		(0.03400)
difference between the two		0.00148		-0.18093***
		(0.00214)		(0.03698)
Growing-season RZSM deficits				
all Legal Amazon	0.00001		0.00128**	
municipalities	(0.00003)		(0.00051)	
high-family-farm		0.00005		0.00158**
municipalities		(0.00003)		(0.00069)
low-family-farm		-0.00005		0.00089*
municipalities		(0.00003)		(0.00051)
difference between the two		0.00010**		0.00069
		(0.00005)		(0.00083)
lagged outcome	YES	YES	YES	YES
controls	YES	YES	YES	YES
fixed effects	YES	YES	YES	YES
$adj.R^2$	0.81	0.81	0.45	0.46
within R^2	0.02	0.02	0.01	0.01
N	184,240	184,030	10,794	10,780

Note: [a]We use the same deforestation and environmental fine measures as in Table B4. But we replace year-round RZSM deficits with two seasonal RZSM deficits, one for the pregrowing season (July-August) and one for the growing season (all other months in the same year). As before, regressions include individual and time fixed effects, lagged dependent variable, and various controls. Standard errors, clustered at the municipality level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C — Mechanism Results Tables

Table C1 — Impacts of Seasonal RZSM Deficits on Rice Production[a]

	area planted		area harvested		output quantity		output revenue		average price	
	average effects	heterogeneous effects	average effects	heterogeneous effects	average effects	heterogeneous effects	average effects	heterogeneous effects	average effects	heterogeneous effects
Pregrowing-season RZSM deficits										
all Legal Amazon	-0.2217		-0.2384		-0.0875		-0.0896		-0.0073	
municipalities	(0.1667)		(0.1668)		(0.1403)		(0.1443)		(0.0175)	
high-family-farm		-0.8927**		-0.9048**		-1.0865***		-1.1114***		-0.0937***
municipalities		(0.3691)		(0.3716)		(0.2587)		(0.3193)		(0.0129)
low-family-farm		-0.1382		-0.1552		-0.0289		-0.0251		-0.0023
municipalities		(0.1509)		(0.1507)		(0.1206)		(0.1249)		(0.0179)
difference between the two		-0.7545*		-0.7497*		-1.0576***		-1.0863***		-0.0913***
		(0.3975)		(0.3997)		(0.2832)		(0.3416)		(0.0174)
Growing-season RZSM deficits										
all Legal Amazon	-0.0010		-0.0003		0.0025		0.0040		0.0011**	
municipalities	(0.0032)		(0.0033)		(0.0035)		(0.0032)		(0.0005)	
high-family-farm		-0.0105**		-0.0099**		-0.0093**		-0.0066		-0.0004
municipalities		(0.0042)		(0.0043)		(0.0037)		(0.0053)		(0.0004)
low-family-farm		0.0045**		0.0053**		0.0071***		0.0075***		0.0017***
municipalities		(0.0020)		(0.0021)		(0.0027)		(0.0029)		(0.0006)
difference between the two		-0.0149***		-0.0152***		-0.0165***		-0.0141**		-0.0021***
		(0.0045)		(0.0046)		(0.0043)		(0.0058)		(0.0006)
lagged outcome	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
controls	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
fixed effects	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
$adj.R^2$	0.85	0.85	0.85	0.85	0.86	0.86	0.85	0.85	0.48	0.48
within R^2	0.43	0.43	0.42	0.43	0.40	0.40	0.40	0.40	0.10	0.10
N^b	6,944	6,944	6,944	6,944	6,944	6,944	6,944	6,944	6,944	6,944

Note: [a]We obtain annual crop production data from the Brazilian Institute of Geography and Statistics and take log for all production measures. Each regression is weighted by the 2005 municipality-specific outcome level. For price regressions, we use 2005 municipality-specific output weights. We deflate revenues and prices using the Broad Producer Price Index (IPA, general) downloaded from IpeaData. [b]There are 496 Legal Amazon municipalities that have consistently-reported rice production data. Standard errors, clustered at the municipality level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C2 — Impacts of Year-Round RZSM Deficits on Cassava Production[a]

	area planted		area harvested		output quantity		output revenue		average price	
	average effects	heterogeneous effects	average effects	heterogeneous effects	average effects	heterogeneous effects	average effects	heterogeneous effects	average effects	heterogeneous effects
all Legal Amazon	-0.0028*		-0.0029*		-0.0034**		-0.0016		0.0021***	
municipalities	(0.0016)		(0.0016)		(0.0015)		(0.0011)		(0.0008)	
high-family-farm		-0.0030		-0.0030		-0.0037*		-0.0020		0.0019*
municipalities		(0.0019)		(0.0020)		(0.0020)		(0.0014)		(0.0011)
low-family-farm		-0.0022		-0.0025		-0.0028*		-0.0004		0.0026***
municipalities		(0.0016)		(0.0016)		(0.0017)		(0.0015)		(0.0007)
difference between the two		-0.0008		-0.0005		-0.0008		-0.0017		-0.0007
		(0.0024)		(0.0024)		(0.0024)		(0.0020)		(0.0012)
lagged outcome	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
controls	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
fixed effects	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
$adj.R^2$	0.93	0.93	0.92	0.92	0.92	0.92	0.86	0.86	0.71	0.71
within R^2	0.47	0.47	0.42	0.42	0.36	0.36	0.36	0.36	0.20	0.20
N^b	9,646	9,646	9,646	9,646	9,646	9,646	9,646	9,646	9,646	9,646

Note: [a]We obtain annual crop production data from the Brazilian Institute of Geography and Statistics and take log for all production measures. RZSM deficit intensity is measured over the full calendar year rather than separate pregrowing/growing windows, reflecting cassava's continuous, multi-year production cycle. As before, each regression is weighted by the 2005 municipality-specific outcome level. For price regressions, we use 2005 municipality-specific output weights. We deflate revenues and prices using the Broad Producer Price Index (IPA, general) downloaded from IpeaData. [b]There are 689 Legal Amazon municipalities that have consistently-reported cassava production data. Standard errors, clustered at the municipality level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C3 — Impacts of Seasonal RZSM Deficits on Soybean Production[a]

	area planted		area harvested		output quantity		output revenue		average price	
	average effects	heterogeneous effects	average effects	heterogeneous effects	average effects	heterogeneous effects	average effects	heterogeneous effects	average effects	heterogeneous effects
Pregrowing-season RZSM deficits										
all Legal Amazon	0.0831		0.0864		0.1343		0.2121		0.0743***	
municipalities	(0.0764)		(0.0762)		(0.1238)		(0.1412)		(0.0281)	
high-family-farm		-0.1368***		-0.1363***		-0.0840***		-0.0030		0.0556***
municipalities		(0.0082)		(0.0083)		(0.0139)		(0.0183)		(0.0087)
low-family-farm		0.1545***		0.1587***		0.2011*		0.2847**		0.0800**
municipalities		(0.0353)		(0.0353)		(0.1117)		(0.1332)		(0.0334)
difference between the two		-0.2913***		-0.2950***		-0.2851**		-0.2877**		-0.0243
		(0.0346)		(0.0346)		(0.1106)		(0.1308)		(0.0325)
Growing-season RZSM deficits										
all Legal Amazon	0.0008		0.0007		0.0030		0.0066**		0.0009	
municipalities	(0.0021)		(0.0022)		(0.0025)		(0.0032)		(0.0009)	
high-family-farm		-0.0756***		-0.0756***		-0.0697***		-0.2594***		-0.1726***
municipalities		(0.0052)		(0.0052)		(0.0067)		(0.0112)		(0.0054)
low-family-farm		0.0008		0.0007		0.0030		0.0066**		0.0009
municipalities		(0.0021)		(0.0022)		(0.0025)		(0.0032)		(0.0009)
difference between the two		-0.0764***		-0.0763***		-0.0727***		-0.2660***		-0.1735***
		(0.0057)		(0.0057)		(0.0075)		(0.0120)		(0.0055)
lagged outcome	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
controls	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
fixed effects	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
$adj.R^2$	0.98	0.98	0.98	0.98	0.98	0.98	0.97	0.97	0.83	0.83
within R^2	0.68	0.68	0.68	0.68	0.62	0.62	0.52	0.52	0.04	0.04
N^b	2,338	2,338	2,338	2,338	2,338	2,338	2,338	2,338	2,338	2,338

Note: [a]We obtain annual crop production data from the Brazilian Institute of Geography and Statistics and take log for all production measures. Each regression is weighted by the 2005 municipality-specific outcome level. For price regressions, we use 2005 municipality-specific output weights. We deflate revenues and prices using the Broad Producer Price Index (IPA, general) downloaded from IpeaData. [b]There are 167 Legal Amazon municipalities that have consistently-reported soybean production data. Standard errors, clustered at the municipality level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C4 — Impacts of Year-Round RZSM Deficits on Sugarcane Production[a]

	area planted		area harvested		output quantity		output revenue		average price	
	average effects	heterogeneous effects	average effects	heterogeneous effects	average effects	heterogeneous effects	average effects	heterogeneous effects	average effects	heterogeneous effects
all Legal Amazon	-0.0001		-0.0002		0.0004		0.0004		-0.0004	
municipalities	(0.0006)		(0.0006)		(0.0007)		(0.0005)		(0.0005)	
high-family-farm		-0.0001		-0.0002		0.0003		0.0004		-0.0001
municipalities		(0.0006)		(0.0006)		(0.0006)		(0.0004)		(0.0003)
low-family-farm		0.0001		-0.0001		0.0012		-0.0022*		-0.0030***
municipalities		(0.0011)		(0.0011)		(0.0012)		(0.0013)		(0.0008)
difference between the two		-0.0002		-0.0001		-0.0009		0.0026**		0.0030***
		(0.0008)		(0.0008)		(0.0009)		(0.0010)		(0.0008)
lagged outcome	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
controls	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
fixed effects	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
$adj.R^2$	0.96	0.96	0.96	0.96	0.95	0.95	0.94	0.94	0.80	0.80
within R^2	0.51	0.51	0.48	0.48	0.51	0.51	0.55	0.55	0.41	0.41
N^b	2,492	2,492	2,492	2,492	2,492	2,492	2,492	2,492	2,492	2,492

Note: [a]We obtain annual crop production data from the Brazilian Institute of Geography and Statistics and take log for all production measures. RZSM deficit intensity is measured over the full calendar year rather than separate pregrowing/growing windows, reflecting sugarcane's continuous, multi-year production cycle. As before, each regression is weighted by the 2005 municipality-specific outcome level. For price regressions, we use 2005 municipality-specific output weights. We deflate revenues and prices using the Broad Producer Price Index (IPA, general) downloaded from IpeaData. [b]There are 178 Legal Amazon municipalities that have consistently-reported sugarcane production data. Standard errors, clustered at the municipality level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C5 — Impacts of Year-Round RZSM Deficits on Cattle Production[a]

	average effects	heterogeneous effects
all Legal Amazon municipalities	-0.00057***	
	(0.00016)	
high-family-farm municipalities		-0.00023
		(0.00031)
low-family-farm municipalities		-0.00069***
		(0.00018)
difference between the two		0.00046
		(0.00035)
one lag of the outcome variable	YES	YES
one lag of forest stock (*asinh*)	YES	YES
price controls	YES	YES
policy controls	YES	YES
year fixed effects	YES	YES
municipality fixed effects[e]	YES	YES
$adj.R^2$	0.98	0.98
within R^2	0.30	0.30
N	10,738	10,738

Note: [a]We obtain annual cattle head count data from the Brazilian Institute of Geography and Statistics. We focus on 767 Legal Amazon municipalities that have consistently reported positive cattle production across years and take log of cattle head counts. Our regression is weighted by the 2005 municipality-specific cattle head count. Standard errors, clustered at the municipality level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C6 — Impacts of Pregrowing-Season RZSM Deficits on Cassava Production in Rice-Cassava Municipalities[a]

	area planted		area harvested		output quantity		output revenue		average price	
	average effects	heterogeneous effects	average effects	heterogeneous effects	average effects	heterogeneous effects	average effects	heterogeneous effects	average effects	heterogeneous effects
Pregrowing-season RZSM deficits[b]										
all Legal Amazon	0.0350**		0.0620***		0.0460***		0.0400**		0.0096	
municipalities	(0.0147)		(0.0135)		(0.0116)		(0.0197)		(0.0187)	
high-family-farm		0.0032		0.0435***		0.0338**		0.0171		-0.0146
municipalities		(0.0088)		(0.0114)		(0.0161)		(0.0152)		(0.0146)
low-family-farm		0.0498***		0.0700***		0.0493***		0.0651***		0.0181
municipalities		(0.0128)		(0.0168)		(0.0157)		(0.0230)		(0.0231)
difference between the two		-0.0466***		-0.0266		-0.0155		-0.0480**		-0.0327
		(0.0151)		(0.0200)		(0.0231)		(0.0243)		(0.0252)
year-round RZSM deficits[c]	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
lagged outcome	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
controls	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
fixed effects	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
$adj.R^2$	0.94	0.94	0.94	0.94	0.93	0.93	0.88	0.88	0.73	0.73
within R^2	0.49	0.49	0.47	0.47	0.42	0.42	0.35	0.35	0.21	0.21
N[d]	6,496	6,496	6,496	6,496	6,496	6,496	6,496	6,496	6,496	6,496

Note: [a]We obtain annual cassava and rice production data from the Brazilian Institute of Geography and Statistics and take log for all production measures. To estimate potential crop substitution effects, we restrict the sample to municipalities where cassava and rice are jointly cultivated. [b]We define the pregrowing season here as July-August of the current year to see the contemporaneous response of cassava production to pregrowing-season drought shock. [c]We also interact year-round RZSM deficits with the high-family-farm municipality group dummy, as in Table C2. [d]There are 464 Legal Amazon municipalities that have consistently-reported cassava production data and produced both cassava and rice. Standard errors, clustered at the municipality level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C7 — Impacts of RZSM Deficits on Irrigated Area Share by Crop[a]

	soybean		rice		other temporary crops		sugarcane	
	average effects	heterogeneous effects	average effects	heterogeneous effects	average effects	heterogeneous effects	average effects	heterogeneous effects
Pregrowing-season RZSM deficits[b]								
all Legal Amazon	1.70e-04		2.59e-03		-7.73e-04			
municipalities	(6.24e-04)		(3.01e-03)		(8.98e-04)			
high-family-farm		-8.62e-04**		-5.46e-04		-1.01e-03		
municipalities		(3.31e-04)		(4.92e-04)		(1.13e-03)		
low-family-farm		5.12e-04		3.62e-03		-6.92e-04		
municipalities		(8.35e-04)		(4.09e-03)		(1.08e-03)		
difference between the two		-1.37e-03		-4.16e-03		-3.22e-04		
		(8.75e-04)		(4.12e-03)		(1.44e-03)		
Growing-season RZSM deficits[b]								
all Legal Amazon	-6.55e-06		-3.29e-05		-2.69e-05*			
municipalities	(1.26e-05)		(2.43e-05)		(1.48e-05)			
high-family-farm		-1.06e-05		-3.35e-05		-3.09e-05*		
municipalities		(1.27e-05)		(2.55e-05)		(1.86e-05)		
low-family-farm		1.48e-06		-2.94e-05		-1.95e-05		
municipalities		(1.35e-05)		(2.21e-05)		(1.43e-05)		
difference between the two		-1.21e-05*		-4.18e-06		-1.14e-05		
		(7.16e-06)		(6.92e-06)		(1.82e-05)		
Year-round RZSM deficits								
all Legal Amazon							-3.10e-06	
municipalities							(3.15e-06)	
high-family-farm								-1.50e-06
municipalities								(2.20e-06)
low-family-farm								-5.96e-06
municipalities								(7.11e-06)
difference between the two								4.46e-06
								(6.98e-06)
lagged outcome	YES	YES	YES	YES	YES	YES	YES	YES
controls	YES	YES	YES	YES	YES	YES	YES	YES
fixed effects	YES	YES	YES	YES	YES	YES	YES	YES
adj.R^2	0.97	0.97	1.00	1.00	0.91	0.91	0.87	0.87
within R^2	0.87	0.87	0.98	0.98	0.47	0.47	0.82	0.82
N[c]	10,780	10,780	10,780	10,780	10,780	10,780	10,780	10,780

Note: [a]The dependent variable is irrigated area share of total agricultural establishment area (2006 Agricultural Census), by crop. Regressions include municipality and year fixed effects, lagged dependent variable, and various controls as the other tables above. [b]To capture the contemporaneous responses of temporary crops' irrigation to agricultural drought, we use two seasonal RZSM deficits within the same year with the pregrowing season from July to August and the growing season covering all other months in the same year. [c]770 municipalities are matched with agricultural establishment data; the number of observations is 10,780. Standard errors, clustered at the municipality level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C8 — Impacts of Year-Round RZSM Deficits on Rural Credit Usage[a]

	crop		cattle		PRONAF[b]		Proirriga[b]
	amounts	contracts	amounts	contracts	amounts	contracts	dummy
Average effects							
all Legal Amazon	-0.0060**	-0.0052***	-0.0082***	-0.0056***	-0.0012	-0.0021	1.42e-06
municipalities	(0.0030)	(0.0015)	(0.0032)	(0.0015)	(0.0025)	(0.0015)	(0.0001)
Heterogeneous effects							
high-family-farm	-0.0094**	-0.0078***	-0.0122***	-0.0094***	-0.0044*	-0.0038*	-0.0001
municipalities	(0.0037)	(0.0022)	(0.0045)	(0.0018)	(0.0027)	(0.0020)	(0.0001)
low-family-farm	0.0000	-0.0007	-0.0009	0.0013	0.0044	0.0009	0.0001
municipalities	(0.0050)	(0.0016)	(0.0019)	(0.0014)	(0.0044)	(0.0015)	(0.0002)
difference	-0.0094	-0.0071***	-0.0113**	-0.0107***	-0.0089*	-0.0047**	-0.0002
	(0.0061)	(0.0026)	(0.0048)	(0.0022)	(0.0050)	(0.0024)	(0.0002)
lagged outcome	YES	YES	YES	YES	YES	YES	YES
controls	YES	YES	YES	YES	YES	YES	YES
fixed effects	YES	YES	YES	YES	YES	YES	YES
N[c]	8,283	8,283	8,283	8,283	3,845	3,845	3,845

Note: [a]Credit amounts and number of contracts are obtained from the Central Bank of Brazil's Matriz de Dados do Crédito Rural (MDCR) for the years of 2014-2019, supplemented by the cattle and crop credit data of 2005-2011 from Assunção et al. (2020). To align with Assunção et al. (2020), I calculate annual credit measures for 2014-2019 based on the same year-round time window from last August to this July, which is almost aligned with our temporary-crop-year window from last July to this June. We deflate credit amounts using the National Consumer Price Index (IPCA, general) downloaded from IpeaData. We take *asinh* to deal with zero values in credit amounts or number of contracts. Regressions include municipality and year fixed effects, lagged dependent variable, and various controls as the other tables above. [b]PRONAF is the main federal credit program for small-scale or family farmers and Proirriga is an investment credit line focused specifically on irrigation and climate-control infrastructure. [c]The number of municipality-year observations is larger for the crop and cattle regressions because PRONAF and Proirriga data are only available for 2014-2019. The heterogeneous-effect regressions for PRONAF and Proirriga use 3,840 observations because one municipality lacks family-farm data. Standard errors, clustered at the municipality level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C9 — The Impacts of Seasonal RZSM Deficits on Rural Credit Usage[a]

	crop		PRONAF		Proirriga
	amounts	contracts	amounts	contracts	dummy
Pregrowing-season RZSM deficits					
all Legal Amazon	0.0277	0.0025	-0.0938	-0.2302**	-0.0188
municipalities	(0.1614)	(0.0824)	(0.1422)	(0.1008)	(0.0128)
high-family-farm	-0.0214	-0.0776	0.2479	0.0869	5.86e-16
municipalities	(0.0327)	(0.0749)	(0.1959)	(0.0787)	(4.51e-16)
low-family-farm	0.0418	0.0267	-0.2014	-0.3290***	-1.87e-16***
municipalities	(0.2116)	(0.1020)	(0.1494)	(0.1059)	(6.69e-17)
difference	-0.0632	-0.1043	0.4492*	0.4159***	7.74e-16*
	(0.2118)	(0.1258)	(0.2458)	(0.1312)	(4.56e-16)
Growing-season RZSM deficits					
all Legal Amazon	-0.0060**	-0.0052***	-0.0012	-0.0021	4.51e-06
municipalities	(0.0030)	(0.0015)	(0.0025)	(0.0015)	(0.0001)
high-family-farm	-0.0094**	-0.0078***	-0.0044*	-0.0038*	4.35e-19
municipalities	(0.0037)	(0.0022)	(0.0027)	(0.0020)	(3.08e-19)
low-family-farm	0.0000	-0.0007	0.0045	0.0010	-9.89e-19
municipalities	(0.0050)	(0.0016)	(0.0043)	(0.0015)	(9.55e-19)
difference	-0.0094	-0.0071***	-0.0089*	-0.0048**	1.42e-18
	(0.0061)	(0.0026)	(0.0050)	(0.0024)	(9.68e-19)
lagged outcome	YES	YES	YES	YES	YES
controls	YES	YES	YES	YES	YES
fixed effects	YES	YES	YES	YES	YES
N[b]	8,283	8,283	3,845	3,845	3,845

Note: [a]We use the same regressions as Table C8 except that we replace year-round RZSM deficits with pregrowing- and growing-season RZSM deficits and drop cattle credit. [b]The heterogeneous-effect regressions for PRONAF and Proirriga use 3,840 observations because one municipality lacks family-farm data. Standard errors, clustered at the municipality level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D — Main Figures

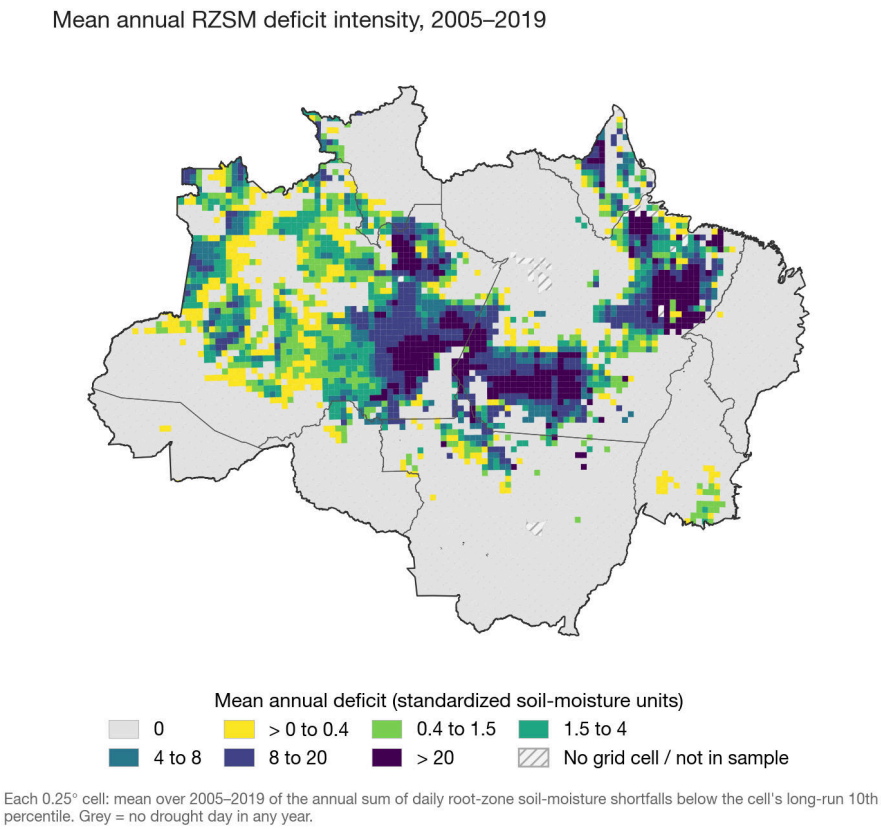


Figure D1 — Mean Annual Drought Intensity, 2005–2019. Grid-level cumulative deficits of standardized daily root-zone soil moisture relative to its long-run bottom-decile threshold. For each grid, we calculate annual deficits by summing the positive differences between the latter and the former across days within a year. Darker grids experienced more severe and/or more prolonged soil-moisture stress over the study period.

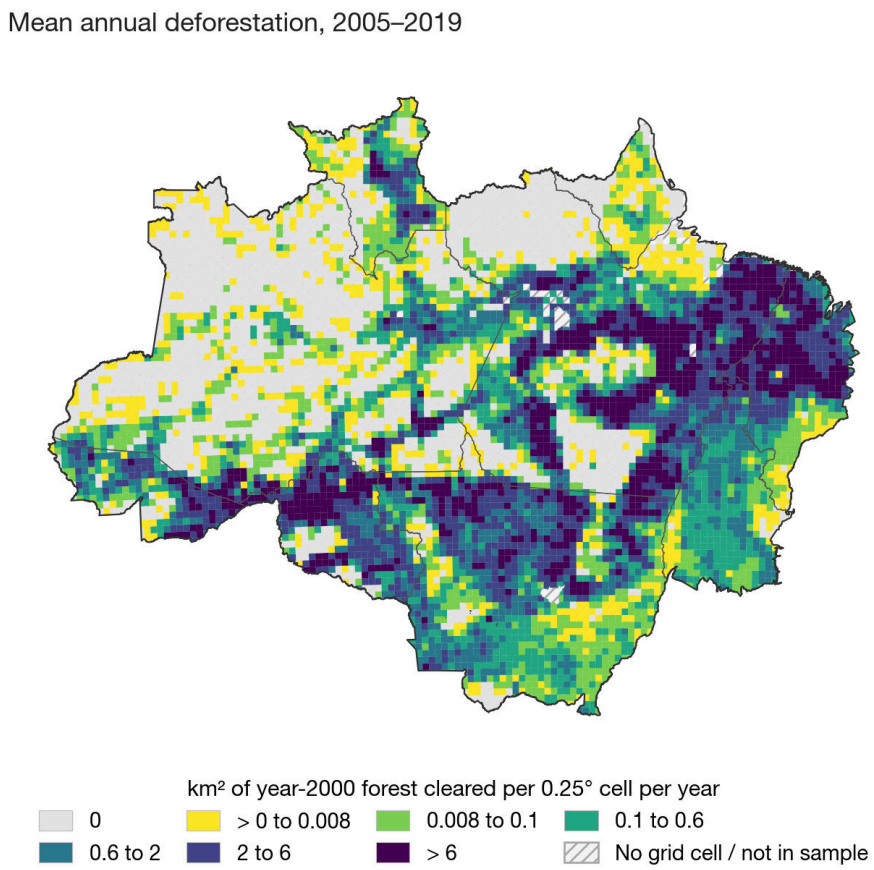


Figure D2 — Mean Annual Deforestation, 2005–2019. Grid-level deforestation in km2 relative to its baseline forest in 2000. For each grid, we calculate annual deforestation by summing areas with baseline forest converted to anthropic land use. Darker grids experienced more deforestation over the study period.

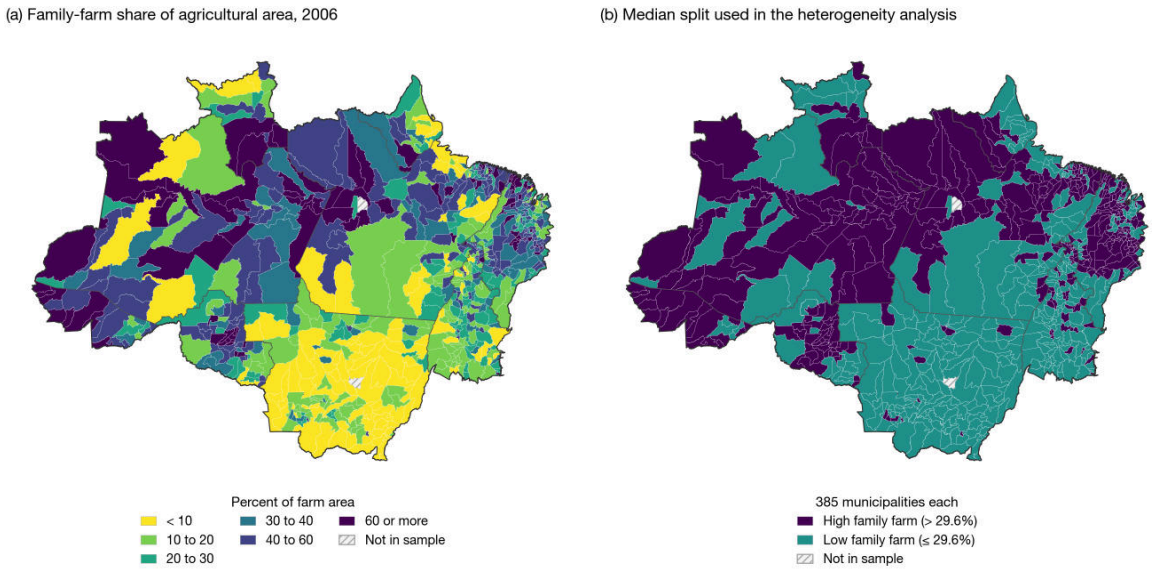


Figure D3 — High- and Low-Family-Farm Municipalities across the Legal Amazon. Municipalities are split at the 2006 Agricultural Census median share of agricultural area in family-farm establishments. High-family-farm municipalities have family-farm area shares above the median (about 30%) while low-family-farm municipalities have family-farm area shares below the median.

E — Robustness Checks Results Tables

Table E1 — The Association Between Local Deforestation and Rainfall[a]

	coefficient	clustered standard errors[b]		
	estimates	municipality	Conley (200 km)	Conley (500 km)
local deforestation	-0.0028	(0.0024)	(0.0031)	(0.0046)
one lag of local forest stock	0.0026	(0.0137)	(0.0252)	(0.0352)
one lag of local rainfall	0.0605	(0.0220)	(0.0246)	(0.0415)

Note: [a]We obtain monthly precipitation data (1/24th degree) from TerraClimate. We calculate area-weighted mean monthly precipitation at our municipality-RZSM grid level before aggregating them across months in a year. There are 11,939 grids in 771 Legal Amazon municipalities that are matched. This regression includes municipality-by-RZSM-grid and year fixed effects plus controls, same as the regression for the average deforestation effect of year-round RZSM deficits. [b]We report the standard errors clustered at the municipality level and Conley spatial clustered standard errors at 200-km and 500-km distance.

Table E2 — Impacts of Alternative RZSM-Based Drought Intensities on Deforestation and Forest-Related Environmental Fines[a]

	deforestation		fine amounts		fine counts	
	avg. effects	hetero. effects	avg. effects	hetero. effects	avg. effects	hetero. effects
Number of days with daily RZSM below its 10th percentile						
all Legal Amazon	0.0004		0.0294**		0.0114**	
municipalities	(0.0003)		(0.0134)		(0.0051)	
high-family-farm		0.0010***		0.0350**		0.0129**
municipalities		(0.0004)		(0.0172)		(0.0065)
low-family-farm		-0.0005		0.0183		0.0085
municipalities		(0.0004)		(0.0172)		(0.0070)
difference between the two		0.0015***		0.0167		0.0044
		(0.0005)		(0.0233)		(0.0093)
RZSM deficits over drought spells lasting 7+ days						
all Legal Amazon	0.00008		0.0082***		0.0035***	
municipalities	(0.00008)		(0.0032)		(0.0012)	
high-family-farm		0.00022**		0.0103**		0.0046***
municipalities		(0.00009)		(0.0041)		(0.0016)
low-family-farm		-0.00011		0.0044		0.0017
municipalities		(0.00010)		(0.0044)		(0.0014)
difference between the two		0.00033***		0.0059		0.0029
		(0.00012)		(0.0058)		(0.0021)
lagged outcome	YES	YES	YES	YES	YES	YES
controls	YES	YES	YES	YES	YES	YES
fixed effects	YES	YES	YES	YES	YES	YES

Note: [a]We use the same deforestation and environmental fine measures as in Table B1 and B3. But we replace year-round RZSM deficits with two alternative drought intensity measures, the number of days with daily RZSM below its 10th percentile and RZSM deficits over drought spells lasting at least 7 consecutive days. As before, regressions include individual and time fixed effects, lagged dependent variable, and various controls. Standard errors, clustered at the municipality level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E3 — Impacts of Year-Round RZSM Deficits on Deforestation and Forest-Related Environmental Fines with Continuous Family-Farm Area Shares[a]

	deforestation	fine amounts	fine counts
Municipalities with 0% family-farm area	-0.00008	0.0034	0.0016
(fully commercial farms)	(0.00011)	(0.0049)	(0.0016)
Municipalities with 100% family-farm area	0.00041**	0.0169	0.0068
(fully family farms)	(0.00020)	(0.0111)	(0.0044)
Marginal effect of family-farm share	0.00049*	0.0135	0.0052
(0% → 100%)	(0.00027)	(0.0148)	(0.0057)
lagged outcome	YES	YES	YES
controls	YES	YES	YES
fixed effects	YES	YES	YES
$adj.R^2$	0.89	0.63	0.69
within R^2	0.29	0.01	0.03
N	184,030	10,780	10,780

Note: [a]We use the same deforestation and environmental fine measures as in Table B1 and B3. Family-farm exposure is measured continuously as the municipality's family-farm area share (0–1), interacted with year-round RZSM deficit intensity. Rows report the estimated effect at 0% and 100% family-farm share (via `lincom`), and the marginal effect of moving across the full range. Regressions include individual and time fixed effects, lagged dependent variable, and various controls as the other tables above. Standard errors, clustered at the municipality level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E4 — Impacts of Year-Round RZSM Deficits on Deforestation: Family-Farm and Rural-Poor-Household Heterogeneity[a]

	(1) family farm	(2) poor household	(3) interaction
Municipalities with fewer family farms	-0.00012		
	(0.00010)		
Municipalities with more family farms	0.00022**		
	(0.00009)		
Municipalities with fewer poor households		-0.00000	
		(0.00010)	
Municipalities with more poor households		0.00015	
		(0.00010)	
Fewer family farms, fewer poor households			-0.00013
			(0.00012)
More family farms, fewer poor households			0.00023**
			(0.00009)
Fewer family farms, more poor households			-0.00008
			(0.00008)
More family farms, more poor households			0.00021*
			(0.00011)
lagged outcome	YES	YES	YES
controls	YES	YES	YES
fixed effects	YES	YES	YES
$adj.R^2$	0.88	0.88	0.88
within R^2	0.25	0.25	0.25
N^b	180,726	180,726	180,726

Note: [a]We obtain the rural household income distribution data (Tabela 1993) in the 2000 Demographic Census from the Brazilian Institute of Geography and Statistics. The dataset records municipality-level household counts by monthly income bins relative to the minimum monthly wage rate. We classify a rural household with its monthly income less than the minimum wage as a poor household, we define municipalities with rural-poor-household shares above the median, roughly 40%, as those with more rural poor households. Column (1) interacts year-round RZSM deficit intensity with the more-family-farm municipality indicator only; Column (2) interacts it with the more-poor-household municipality indicator only; Column (3) includes both indicators and their triple interaction with year-round RZSM deficit intensity. Reported coefficients are linear combinations giving the estimated effect for each named subgroup. Regressions include municipality-RZSM grid and year fixed effects, lagged dependent variable, and various controls as the other tables above. [b]We are only able to match the rural household income distribution data with 755 Legal Amazon municipalities in our main sample. So, the number of observations slightly drops. Standard errors, clustered at the municipality level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E5 — Impacts of Year-Round RZSM Deficits on Deforestation: Family-Farm and Public-Land Heterogeneity[a]

	(1) family farm	(2) public land	(3) interaction
Municipalities with fewer family farms	-0.00012		
	(0.00010)		
Municipalities with more family farms	0.00022**		
	(0.00009)		
Municipalities with less public land		-0.00019	
		(0.00017)	
Municipalities with more public land		0.00009	
		(0.00008)	
Fewer family farms, less public land			-0.00029
			(0.00021)
More family farms, less public land			0.00015
			(0.00026)
Fewer family farms, more public land			-0.00010
			(0.00010)
More family farms, more public land			0.00023**
			(0.00009)
lagged outcome	YES	YES	YES
controls	YES	YES	YES
fixed effects	YES	YES	YES
$adj.R^2$	0.87	0.87	0.87
within R^2	0.25	0.25	0.25
N	184,030	184,030	184,030

Note: [a]We obtain the public land data from the CNFP (Cadastro Nacional de Florestas Públicas, 2024 edition, rectified July 2025). The dataset includes public land (mostly forest) that is either designated or undesignated (not yet allocated to a specific use). We focus on undesignated public land that is less likely to be protected. We classify municipalities undesignated public land area shares above the median as municipalities with more public land. Column (1) interacts year-round RZSM deficit intensity with the more-family-farm municipality indicator only; Column (2) interacts it with the more-public-land municipality indicator only; Column (3) includes both indicators and their triple interaction with year-round RZSM deficit intensity. Reported coefficients are linear combinations giving the estimated effect for each named subgroup. Regressions include municipality-RZSM grid and year fixed effects, lagged dependent variable, and various controls as the other tables above. Standard errors, clustered at the municipality level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F — Carbon Valuation of Drought-Induced Deforestation

This appendix translates the estimated effect of drought on deforestation (Section 6, Table B1) into a monetary valuation of the associated carbon-emission externality. The goal is to express the marginal social damage of drought-induced deforestation in dollar terms, providing a policy-relevant benchmark against which the cost of the rural finance interventions discussed in the main text can be compared. As explained in the main text, we focus on the additional drought-induced deforestation in high-family-farm municipalities that may be *avoidable* through policy interventions. The valuation proceeds in three steps: (i) recovering the physical forest loss in hectares implied by the estimated asinh-scale deforestation response; (ii) converting hectares cleared into committed gross carbon-dioxide emissions; and (iii) applying a range of social-cost-of-carbon (SCC) values to monetize those emissions. We treat the resulting figures as a lower bound on total social damage, since they omit belowground and soil carbon, co-emitted greenhouse gases from associated burning, and non-carbon ecosystem-service losses.

F.1. From the asinh deforestation response to hectares cleared

Our deforestation outcome is the inverse hyperbolic sine of absolute annual deforestation, $Y_{i,t} = \text{asinh}(h_{i,t})$, where $h_{i,t}$ denotes hectares of deforestation in municipality-RZSM grid cell i in year t. Because the outcome is nonlinear in hectares, the estimated coefficient does not map directly into a level effect; we therefore recover predicted drought-attributable hectares from the fitted model rather than relying on a local approximation. We focus on the heterogeneous-effect specification with regression results reported in Table B1. We aim to back out the additional drought-induced deforestation in high-family-farm municipalities relative to low-family-farm municipalities. Such drought-induced deforestation is potentially avoidable via policy interventions.

Let $\widehat{Y}_{i,t}$ denote the fitted deforestation outcome from the estimated heterogeneous-effect regression model, and let $\widehat{Y}_{i,t}^{\,0}$ denote the corresponding counterfactual fitted value obtained by setting the high-family-farm municipality indicator to zero while holding all other covariates (including RZSM deficits), the lagged outcome (2005 only), and the fixed effects at their observed or estimated values. Back-transforming out of asinh units, drought-induced deforestation in high- and low-family-farm municipalities are

$$\widehat{h}_{i,t} = \sinh\big(\widehat{Y}_{i,t}\big), \qquad \widehat{h}_{i,t}^{\,0} = \sinh\big(\widehat{Y}_{i,t}^{\,0}\big), \tag{F.1}$$

and the avoidable drought-induced deforestation in cell i and year t is the difference

$$\Delta h_{i,t} = \widehat{h}_{i,t} - \widehat{h}_{i,t}^{\,0} = \sinh\big(\widehat{Y}_{i,t}\big) - \sinh\big(\widehat{Y}_{i,t}^{\,0}\big). \tag{F.2}$$

Aggregating over the cells and years of interest yields the total avoidable drought-attributable

hectares cleared,

$$\Delta H = \sum_{i \in \mathcal{S}} \sum_{t} \Delta h_{i,t}, \tag{F.3}$$

where $\mathcal{S}$ is the set of cells over which the valuation is conducted—for example, all cells or the cells exposed to a specific drought episode such as 2016.

F.2. From hectares to committed carbon-dioxide emissions

We convert cleared area into committed gross CO_2 emissions following the standard biomass-accounting approach. Aboveground biomass (AGB) is converted to carbon using the IPCC default carbon fraction $\phi = 0.47$ for tropical forest (IPCC 2019), and carbon is converted to carbon dioxide using the molecular-weight ratio $44/12 \approx 3.67$. Treating all aboveground carbon on cleared land as committed emissions, the per-hectare emission factor is

$$EF = \text{AGB} \times \phi \times \tfrac{44}{12} \quad (\text{t CO}_2\,\text{ha}^{-1}). \tag{F.4}$$

For the Brazilian Amazon we adopt the LiDAR-calibrated basin-average biomass of AGB = 174 $\text{t}\,\text{ha}^{-1}$ from Ometto et al. (2023), equivalent to roughly 82 $\text{t C}\,\text{ha}^{-1}$ and an emission factor of about 300 $\text{t CO}_2\,\text{ha}^{-1}$. This is a deliberately conservative central value: it counts only aboveground live biomass and excludes belowground biomass, deadwood, litter, and soil organic carbon, which can add substantially to total committed emissions where clearing is followed by burning or where organic soils are involved. Total committed emissions from the avoidable drought-induced deforestation are then

$$E = \Delta H \times EF. \tag{F.5}$$

Table F1 reports EF under low, central, and high carbon-density assumptions spanning the range reported for Amazonian forest.

Table F1 — Committed gross CO_2 emission factor per hectare of forest cleared

Carbon-density scenario	AGB (t/ha)	Carbon (t C/ha)	EF (t CO_2/ha)
Low (degraded / transitional)	110	52	190
Central (basin average, Ometto et al. 2023)	174	82	300
High (dense primary forest)	300	141	517

Notes: Carbon fraction $\phi = 0.47$ (IPCC 2019); CO_2/C ratio $44/12 \approx 3.67$. Figures rounded. Estimates count aboveground live biomass only and therefore understate total committed emissions.

F.3. Monetizing emissions with the social cost of carbon

The dollar value of the carbon externality from the avoidable drought-induced deforestation is the product of committed emissions and the social cost of carbon,

$$D = E \times \text{SCC} = \Delta H \times EF \times \text{SCC}. \tag{F.6}$$

There is no single agreed SCC, and the choice materially affects the valuation, so we report a range that brackets the values most used in policy and academic work. Our central, policy-relevant benchmark is the U.S. Environmental Protection Agency's official schedule (U.S. Environmental Protection Agency 2023), which uses a modular framework with dynamic Ramsey discounting calibrated to three near-term rates. For a 2020 emissions pulse, the EPA central SC-CO_2 is \$190 per metric ton at the 2.0% near-term discount rate, with the discount-rate range spanning \$120 (2.5%) to \$340 (1.5%). For comparison we report two lower integrated-assessment-model benchmarks—the U.S. interagency working group interim central estimate of about \$51 per ton (Interagency Working Group on Social Cost of Greenhouse Gases 2021), and Nordhaus's DICE estimates (\$31 per ton for 2015 emissions in DICE-2016R, Nordhaus 2017; \$50 per ton on the cost–benefit-optimal path and \$66 baseline for 2020 in DICE-2023, Barrage and Nordhaus 2024)—and Pindyck's survey-based estimates as an independent, non-IAM benchmark, which yield an average SCC of roughly \$80–\$100 per ton after trimming outliers and \$200 per ton or higher for the untrimmed expert distribution (Pindyck 2019). These benchmarks are summarized in Table F2.

Table F2 — Social-cost-of-carbon benchmarks (2020 dollars per metric ton CO_2)

Source	SCC (\$/t CO_2)	Basis
Nordhaus (Nordhaus 2017), DICE-2016R	31	2015 emissions (2010 US\$); baseline path
IWG (Interagency Working Group on Social Cost of Greenhouse Gases 2021) interim	51	2020 emissions; 3% constant discount rate
Barrage and Nordhaus (Barrage and Nordhaus 2024), DICE-2023	50–66	2020 emissions; \$50 optimal, \$66 baseline
Pindyck (Pindyck 2019), trimmed	80–100	Expert survey with outliers trimmed
EPA (U.S. Environmental Protection Agency 2023) central	190	2020 emissions; 2.0% near-term Ramsey discount rate
EPA (U.S. Environmental Protection Agency 2023), 1.5% rate	340	2020 emissions; 1.5% near-term Ramsey discount rate
Pindyck (Pindyck 2019), untrimmed	≥200	Full expert survey distribution

Notes: Estimates use different base years and discount rates and are not inflation-harmonized. Values should be converted to a common base year in the final draft. IAM = integrated assessment model.

F.4. Illustrative valuation and sensitivity

To make the magnitudes concrete, Table F3 applies Equation (F.6) to the illustrative placeholder of $\Delta H = 10{,}000\ \mathrm{ha\,yr^{-1}}$ of drought-attributable forest loss (to be replaced with the sample-based estimate from Equation (F.3)). At the central carbon density of $300\ \mathrm{t\,CO_2\,ha^{-1}}$, 10,000 ha implies 3.0 $\mathrm{Mt\,CO_2}$ of committed emissions, valued at about \$570 million at the EPA central SCC of \$190, and ranging from roughly \$93 million at Nordhaus's \$31 figure to about \$1.0 billion at the EPA \$340 value. Because the valuation is linear in

ΔH, the entire table rescales by the single factor (ΔH/10,000) once the sample-based hectare estimate is inserted.

Table F3 — Annual carbon-damage value of 10,000 ha of drought-induced clearing (US$ millions)

	SCC ($/t CO_2)			
Emission factor	$31	$51	$190	$340
Low (190 t CO_2/ha)	59	97	361	646
Central (300 t CO_2/ha)	93	153	570	1,020
High (517 t CO_2/ha)	160	264	982	1,758

Notes: Cell entries are $\Delta H \times EF \times SCC$ in millions of US$, for $\Delta H = 10{,}000$ ha yr^{-1} (illustrative placeholder). Damages scale linearly in ΔH; rescale by (ΔH/10,000) for the sample-based estimate.

The spread across the table shows that the carbon-density assumption and the SCC choice are the dominant sources of uncertainty—together they move the valuation by roughly an order of magnitude—whereas the estimated deforestation response enters only through ΔH. We therefore report the central combination alongside the full range rather than a single point estimate.

F.5. Interpretation and link to policy

The valuation should be read as the marginal external carbon cost of the deforestation that drought induces, concentrated—as the main results show—among family-farm municipalities. Because this clearing appears to function as a short-run coping response to drought-induced income shocks rather than as forward-looking commercial land conversion, the carbon damages estimated here are, in principle, avoidable through interventions that relax the underlying liquidity constraint. The relevant policy comparison is therefore between the per-ton carbon damage computed above and the per-ton cost of providing the rural insurance or credit access that would blunt the coping motive. If the cost of such programs per ton of avoided CO_2 lies below the SCC range in Table F2, they are justified on carbon grounds alone, before accounting for their direct welfare benefits to vulnerable smallholders.